Available online at www.sciencedirect.com
SCIENCE DIRECT
Precambrian Research 147 (2006) 234–259
Precambrian Research
www.elsevier.com/locate/precamres

A new paleomagnetic pole for the Neoproterozoic Uinta Mountain supergroup, Central Rocky Mountain States, USA

Arlo B. Weil [a,*], John W. Geissman [b], James M. Ashby [b]

[a] *Bryn Mawr College, Department of Geology, Bryn Mawr, PA 19010, United States*
[b] *University of New Mexico, Department of Earth and Planetary Sciences, Albuquerque, NM 87131, United States*

Received 17 November 2004; received in revised form 26 May 2005; accepted 6 January 2006

Abstract

Hematite-cemented strata of the Neoproterozoic age (~800–750 Ma) Uinta Mountain Group (UMG) provide a new well-defined mid-Neoproterozoic paleomagnetic pole for Laurentia. Thermal demagnetization results and rock magnetic experiments reveal the presence of two discrete magnetizations in UMG rocks. Both magnetizations are carried by hematite. The dominant characteristic remanent magnetization (ChRM) is an east-directed (or antipode) shallow, positive to negative inclination magnetization that is typically unblocked over a narrow range of high laboratory temperatures between 660 and 680 °C. The second magnetization, exclusively of normal polarity, is north to northeast-directed and has a moderate to steep, positive inclination and probably represents a recent or modern day viscous remanent magnetization. The response of UMG samples to thermal and chemical demagnetization, comparison with previously obtained data from UMG rocks in the area, results from rock magnetic experiments, and the inferred, low-latitude of Laurentia during UMG deposition in the Neoproterozoic are all consistent with the interpretation that the west or east-directed, shallow magnetization is a near primary remanence. The presence of dual polarities of the ChRM, with the west-directed remanence inferred to be of normal polarity and the east-directed of reverse polarity, suggests that magnetization acquisition in the UMG rocks took place over a sufficient amount of time to average secular variation of the geomagnetic field. The mean paleomagnetic pole derived from the inferred primary magnetizations from UMG strata (0.8°N, 161.3°E, $\alpha_{95} = 4.6°$, $N = 9$ sampling localities consisting of 79 sites) is fully consistent with recently reported results from rocks of mid-Neoproterozoic age for southwest Laurentia, which define a tight counterclockwise loop, located in the south Pacific, in the apparent polar wander path for Laurentia during the Neoproterozoic.
© 2006 Elsevier B.V. All rights reserved.

Keywords: Neoproterozoic; Tectonics; Uinta Mountain Group; Rodinia; Paleomagnetism

1. Introduction

Fundamental to our understanding of the Late Proterozoic (Neoproterozoic) tectonic history and life cycle of the Rodinia supercontinent is a complete understanding of Laurentia's paleogeography during the time period between about 1200 and 550 Ma. Despite considerable uncertainty on the exact configuration of Rodinia (Meert and Powell, 2002; Meert and Torsvik, 2003; Pisarevsky et al., 2003), it is well documented that Laurentia is surrounded by Neoproterozoic rift margins (Bond and Kominz, 1984) and that collision belts with similar ages are found throughout Gondwana (Meert, 2003). Consequently, whatever the definitive word turns out to be with respect to the paleogeography and longevity of

* Corresponding author. Tel.: +1 610 526 5113; fax: +1 610 526 5086.
E-mail address: aweil@brynmawr.edu (A.B. Weil).

0301-9268/$ – see front matter © 2006 Elsevier B.V. All rights reserved.
doi:10.1016/j.precamres.2006.01.017

Rodinia, some manifestation of a supercontinent must have existed during the Neoproterozoic, and this supercontinent later broke up to form the daughter supercontinent Gondwana by 550 Ma (Meert, 2003; Meert and Torsvik, 2003). In the absence of preserved oceanic lithosphere and marine magnetic anomaly records for any time prior to the Jurassic, paleomagnetic data provide the only quantitative means to infer ancient continental paleogeography. However, the current Proterozoic paleomagnetic database is only sufficient enough to test paleogeographic relationships between two or three cratons at very discrete time intervals (Meert and Torsvik, 2003). For Laurentia, the centrally positioned craton in Rodinia, there remains a paucity of high-quality paleomagnetic data for much of the Neoproterozoic, and additional data are sorely needed to define the paleogeographic and tectonic history of Laurentia throughout the lifecycle of Rodinia.

The current paucity of well-dated Neoproterozoic paleomagnetic poles for Laurentia and surrounding cratons has resulted in ambiguity about the paleogeography of Laurentia for the key period of Proterozoic time in which the Rodinia supercontinent is thought to have undergone final amalgamation and ultimately dispersal (Roy, 1983; Park, 1994; Buchan et al., 2000; Meert and Powell, 2002). This sparse dataset is in stark contrast to some intervals during the latest part of the Mesoproterozoic (we use the International Commission on Stratigraphy boundary of 1.0 Ga for the Mesoproterozoic and Neoproterozoic, e.g., Gradstein et al., 2005) for Laurentia (e.g., the Logan Loop), which have the most complete Precambrian apparent polar wander paths (APWP) of any of the major continents. The Logan Loop, the most important of these well-defined Mesoproterozoic swaths, is largely derived from rocks of the Keweenawan rift system of northeastern Laurentia (1.1–1.0 Ga) and is defined by over 60 poles having, overall, well-determined stratigraphy, structure and geochronology (see review in Halls and Pesonen, 1982). On the other hand, the existing poles for the latest Mesoproterozoic and Neoproterozoic, especially the interval between 1050 and 600 Ma, come from predominantly high-grade metamorphic rocks from the Grenville Province of northeastern Laurentia, which have been either chemically and or thermally remagnetized (e.g., McWilliams and Dunlop, 1978; Hyodo and Dunlop, 1993; Alvarez and Dunlop, 1998; Warnock et al., 2000). Although Laurentia's APWP has no substantial spatial discontinuities between the Keweenawan and Grenville pole sequences, the direction and sense of younging of the Grenville aged part of the APWP remains contested (Fig. 1a) (McWilliams and Dunlop, 1978; Berger et al., 1979; Berger and York, 1981; Roy, 1983; Hyodo and Dunlop, 1993; Weil et al., 1998; Walderhaug et al., 1999; Meert and Torsvik, 2003; Pisarevsky et al., 2003). In general, the problem with Laurentia's Proterozoic APWP is that it consists of a few well-defined fragments, which are separated by sections of time as long as 200 Ma that lack high-quality paleomagnetic poles (Fig. 1b) (Irving, 1979; Roy, 1983; Pisarevsky et al., 2003).

The uncertainties associated with Laurentia's APWP in the latest Mesoproterozoic and Neoproterozoic have major implications and consequences for how paleomagnetic-based configurations for Rodinia are con-

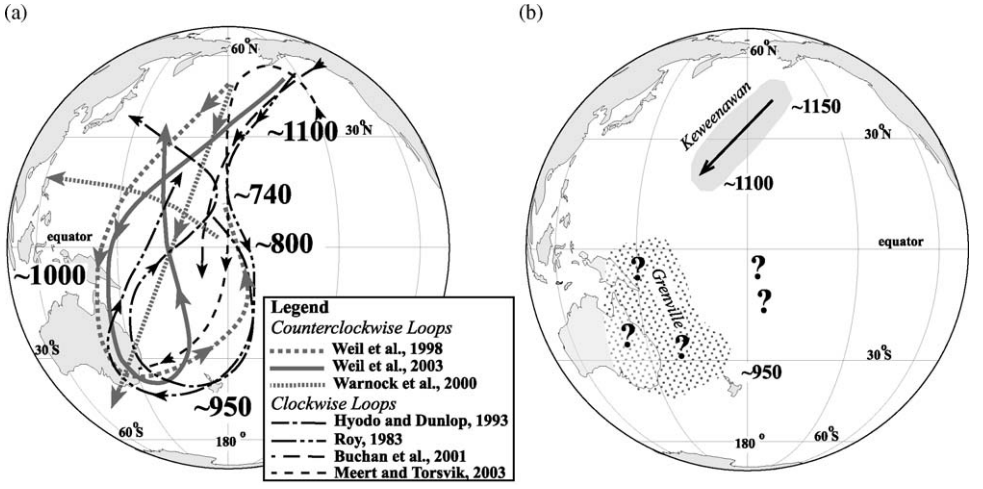


Fig. 1. (a) Representative Late Mesoproterozoic to Neoproterozoic apparent polar wander paths for Laurentia. Notice the combination of both clockwise and counterclockwise loops, as well as differences in the loop width as it passes down through the south Pacific. References for the different paths are given in the figure legend. (b) Grey shaded regions highlight the sections of Laurentia's APWP for the period between 1200 and 550 Ma that are well defined. Hatched area is the general location of the poorly defined poles from metamorphic rocks of the Grenville Province.

structed. The number of well-dated high-quality paleomagnetic poles for the relevant time period is small; using only key poles, it is nearly impossible to base reconstructions on the few results from a given continent for the purposes of paleogeography (Buchan et al., 2001; Meert and Torsvik, 2003). However, if several high-quality poles are available for a given pair of continents, and a clearly defined APWP path can be generated for each block, then APWP geometry becomes a useful tool for generating relative continental reconstructions (e.g., Weil et al., 1998). Consequently, it is imperative that the correct time progression and geometry of Laurentia's APWP be known before further speculation is made on the paleogeography of the Rodinia supercontinent during its lifecycle and ultimately during its rift-to-drift history. At present, there are only narrow time windows for which there is sufficient data to construct robust sections of APWP for purposes of comparison.

The fragmentation of the Rodinia supercontinent between 800 and 550 Ma and its rift-to-drift history are also heavily contested (Powell et al., 1993; Torsvik et al., 1996; Buchan et al., 2000; Torsvik et al., 2001a; Meert and Powell, 2002; Wingate et al., 2002; Pesonen et al., 2003; Torsvik, 2003; Meert and Torsvik, 2003; Wingate and Evans, 2003). This time interval saw, in addition to the fragmentation of an equatorial supercontinent (Hoffman, 1991; Dalziel, 1992), the radiation of eukaryotic organisms (Knoll, 1992), and possibly several dramatic climatic events during which continental glaciation and sea-ice survive at low-latitude (Kaufman and Knoll, 1995; Kaufman, 1997; Hoffman et al., 1998a,b). Kirschvink (1992) proposed, on the basis of limited paleomagnetic data interpreted to indicate that many of these glacial deposits formed at equatorial latitudes, a "snowball" Earth hypothesis (Harland, 1964), during which time the planet is covered in ice, weathering reactions basically cease, and carbon dioxide concentrations gradually increase in the atmosphere, ultimately creating a global, CO_2-induced catastrophic warming. Hoffman et al. (1998a,b) and Hoffman (1999) combined stratigraphic, sedimentologic, and stable-isotope data that strongly support such a hypothesis and in fact argue for multiple episodes of global icehouse conditions and consequent rapid warming during the mid to late Neoproterozoic. With a growing body of information supporting periods of exceedingly unusual climate behavior for Earth in the Neoproterozoic, additional, robust paleomagnetic data for this time period have become even more important. Some initial paleomagnetic data from mid-Neoproterozoic rocks from North America (e.g., Park and Aitken, 1986; Elston, 1993; Park et al., 1995; Weil et al., 2003, 2004) are consistent with the possibility of North America lying at or near the equator during this time period, but some of these data sets are of limited quality and their overall reliability is open to question.

To better understand the drift history and the APWP of Laurentia for the latest Neoproterozoic, we have undertaken a comprehensive paleomagnetic study of the Neoproterozoic Uinta Mountain Group (UMG), northeast Utah, northwest Colorado, and southwest Wyoming. The UMG is one of the thickest and best preserved Neoproterozoic sedimentary rock successions in North America. Because the UMG is one of the few sedimentary successions associated with the incipient dispersal of Rodinia, and because of the excellent preservation of this succession (i.e., lack of burial metamorphism) it provides an ideal focus for detailed paleomagnetic study. The emphasis of this study is on correlating the paleomagnetic data from the UMG succession with those from other, similar aged Neoproterozoic successions along the western Cordillera to better define the late stage history of Rodinia. Specifically these data help better define the geometry and temporal progression of Laurentia's Neoproterozoic APWP.

2. Geologic setting

Much of the Rocky Mountain foreland is characterized by uplifts exposing Precambrian basement rocks that have been displaced along moderate to steeply dipping reverse faults (Gries, 1983; Bird, 1998). The Uinta Mountains are an east–west trending Laramide-style uplift in northeast Utah and northwest Colorado (Fig. 2). The belt is flanked to the north and south by reverse faults that juxtapose Proterozoic UMG rocks over Paleozoic rocks with a total horizontal displacement of about 6 km along the northern bounding fault and 4 km along the southern bounding fault with a total north–south shortening of about 16 km during Laramide deformation (Stone, 1995).

The UMG has been interpreted as an aulacogen, or failed rift arm (Stone, 1995), which opened westward (in present-day coordinates) into an Atlantic-type passive margin (Harrison et al., 1974). However, no Proterozoic mafic volcanic rocks have been found within the section and the UMG basin likely predates the development of the Cordilleran miogeocline by about 200 Ma (Bond and Kominz, 1984). Moreover, the total dimension of the basin westward and its relationship to other Proterozoic basins is unknown (Link et al., 1993). Condie et al. (2001) suggest that the UMG was deposited during rift initiation and subsequent rift-to-drift of East Gondwana cratons away from the present-day Cordilleran margin of Laurentia.

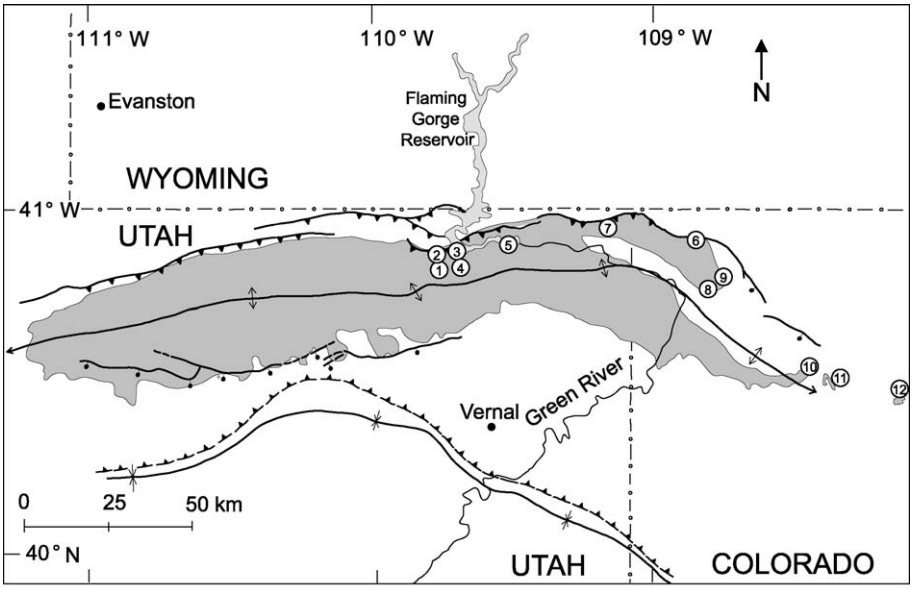


Fig. 2. Generalized map of the Uinta Mountain Group (grey filled area). Paleomagnetic sample areas marked with numbers. (1) Browne Lake, (2) Sheep Creek Canyon, (3) Down Springs, (4) Carter Creek, (5) Flaming Gorge Dam, (6) Sparks–Talamantes Creek, (7) Home Mountain Canyon, (8) Bull Canyon, (9) Irish Canyon, (10) Lone Mountain, (11) Cross Mountain, (12) Juniper Mountain. Base geology after Bresler (1981).

The sedimentary facies within the UMG have been interpreted as dominantly fluvial basin deposits with a source to the east and north (Hansen, 1965; Sanderson, 1984; Sanderson and Wiley, 1986; Farmer and Ball, 1997) or a mixed fluvial-marine environment (Wallace and Crittenden, 1969; Wallace, 1972; Dehler et al., 2006). Detailed sedimentologic and stratigraphic studies are warranted to resolve questions about depositional environments, paleogeography, and basin type.

The UMG ranges in thickness from 4 km in the west to over 7 km in the east (Link et al., 1993). The western UMG is composed of quartzose and arkosic sandstone and interbedded mudstone, much of which is hematite-cemented. The eastern section consists of conglomerate, sandstone and mudstone, and most sandstone units in the east are hematite-cemented redbeds (Hansen, 1965). The UMG nonconformably overlies the Red Creek Quartzite and Owiyukuts Complex (~1.7–~2.7 Ga, Hansen, 1965; Sears et al., 1982). The Red Creek Quartzite and Owiyukuts Complex are composed of metaquartzite, amphibolite and mica schist (Schell, 1969; Sears et al., 1982; Hansen, 1984) and are restricted to the structurally high, northeastern region of the Uinta Mountains. The UMG is separated from the overlying Middle Cambrian strata by an angular unconformity or disconformity (Bryant and Nichols, 1988; Sprinkel et al., 2002).

A mid-Neoproterozoic (750 Ma) age is generally inferred for at least the upper western part of the UMG based on correlation of *Chuaria* and (or) *Melanocyrillium* microfossils in the Red Pine Shale with the Beck Springs Dolomite in the upper part of the Death Valley Pahrump Group and the Chuar Group of the Grand Canyon Supergroup (Vidal and Ford, 1985; Porter and Knoll, 2000; Dehler et al., 2001a,b), from which an U–Pb zircon age determination of 742 ± 6 Ma on an ash layer has been reported (Karlstrom et al., 2000). Recently, U/Th–Pb microprobe dating of an early diagenetic monazite grain has been reported from the Red Pine Shale by Williams et al. (2003) of about 700–800 Ma, which is consistent with recent regional correlations of the UMG with other Neoproterozoic successions (Dehler et al., 2001a,b). Ongoing studies by Dehler et al. (2006) indicate that all UMG facies are genetically related and are all mid Neoproterozoic (800–750 Ma) in age.

Previous paleomagnetic data from UMG rocks are limited. Bresler (1981) reported data from seven sites, four in the western Uinta Mountains and three in the easternmost Uintas (173 samples total). Five sites revealed west-directed and shallow, negative inclination magnetizations and the mean of these data yields a pole of 0.6°N, 157.2°E. A sixth locality (8 accepted of 13 samples) from the western Uintas yielded east-directed and shallow, positive inclination magnetizations with a mean of, in stratigraphic coordinates, Decl. = 106.3°, Incl. = 22.1°, α_{95} = 15.9° (pole of 4.6°N, 140.1°E). Bresler (1981) suggested that the difference between the pole positions derived from data from localities 1 through 5 and locality 6, which was the stratigraphically highest section

sampled, possibly indicated some apparent polar wander during deposition. We note that the pole determined from locality 6 involved only eight samples and that the poles are not statistically significant.

Assuming a primary magnetization in these rocks, the UMG data reported by Bresler (1981) provide a limited age constraint, in that the two poles have an equatorial location and are consistent with ages of either about 1010 Ma or about 800 Ma, based on the relatively poorly known and still controversial mid-Neoproterozoic apparent polar wander (APW) path for Laurentia.

After UMG deposition, no major deformation affected the area prior to Laramide crustal shortening beginning in the latest Cretaceous. Stone (1993) suggested that, based on a comparison with younger rifted continental margins, that it was possible up to 3 km of uplift and erosion beveled the area covered by UMG strata at the end of the Neoproterozoic prior to early Paleozoic sedimentation.

3. Methods

Twelve sampling localities were established in UMG rocks. These include seven localities in the eastern Uintas (Home Mountain Canyon, Irish Canyon, Bull Canyon, Sparks, Lone Mountain, Cross Mountain, and Juniper Mountain) and, five localities in the north-central Uintas (Sheep Creek Canyon, Browne Lake, Carter Creek, Dowd Springs, and Flaming Gorge Reservoir dam) (Fig. 2). The localities were selected based on exposure of continuous and thick sections of strata. Sampling was focused on hematite-cemented medium to fine-grained sandstone and siltstone.

Each site was defined by an individual stratigraphic bed, or several successive beds, and 7–12 independently oriented samples were obtained. A total of 195 sites from 12 localities were collected in UMG strata.

Natural remanent magnetization (NRM) data were obtained using superconducting rock magnetometers in magnetically shielded rooms at the University of New Mexico and at the University of Michigan. Stepwise demagnetization was performed to determine the direction of characteristic remanent magnetizations (ChRM). For all UMG rocks, both thermal and chemical demagnetization, using reagent grade hydrochloric acid at room temperature (Henry, 1979), were used to effectively isolate ChRMs. Chemical demagnetization was performed in a low field space (H ~200 nT) in a fume hood. Isothermal remanent magnetization (IRM) acquisition and backfield demagnetization experiments, using a pulse DC magnet capable of fields of 3.2 T, were performed on selected samples from all localities to further evaluate the magnetic mineralogy. Three-axis IRM thermal demagnetization experiments (Lowrie, 1990) were performed using three DC fields (3.0, 0.3, and 0.03 T; or "hard," "medium" and "soft," respectively) along three orthogonal axes.

Remanence directions were calculated from principal component analysis (Kirschvink, 1980) of linear vectors picked from orthogonal projection demagnetization plots (Zijderveld, 1967). Estimated directions for specimens were accepted for site mean determinations if unanchored maximum angular deviation (MAD) values were less than 15°. Specimens that could not be resolved to MADs of less than 15°, as well as those specimens with anomalous directions (i.e., individual direction greater than two angular standard deviations from the estimated mean) were rejected for site mean calculations. Site means were calculated by averaging the sample directions, using the method of Fisher (1953). Data from 8 of 10 sites taken from UMG rocks at the Bull Canyon (BC) locality required combined remagnetization circle and setpoint analysis (McFadden and McElhinny, 1988) to approximate ChRM directions. Fold tests were performed where possible to characterize the relative age of magnetization acquisition.

4. Results

4.1. Paleomagnetism

Samples from UMG rocks typically contain NRMs that consist of a single, well-defined magnetization component (stable endpoint behavior) that is either, in *in situ* coordinates, from NNE (~035°) to due north-directed with relatively steep positive inclinations (~+70 to 60°) (Fig. 3), or west-directed (~270°) with shallow, negative to positive inclinations (~+10 to −10°) (or its antipode) (Figs. 3 and 4). Some samples carry two components, a relatively low laboratory unblocking temperature component (<~500 °C) that is north and positive, which is superimposed on a higher laboratory unblocking temperature component (fully unblocked by ~680°) of west declination and shallow inclination (or its antipode). The west-directed and shallow inclination remanence (and antipode) is the principal magnetization observed in over 50% of our UMG collection and, as discussed further below, is interpreted as the primary magnetization in UMG strata. Notably, a considerable fraction of the collection (~45%) is characterized by a single normal polarity magnetization component (Figs. 3 and 5) that, on the basis of its direction, must be young (less than mid-Cretaceous in age).

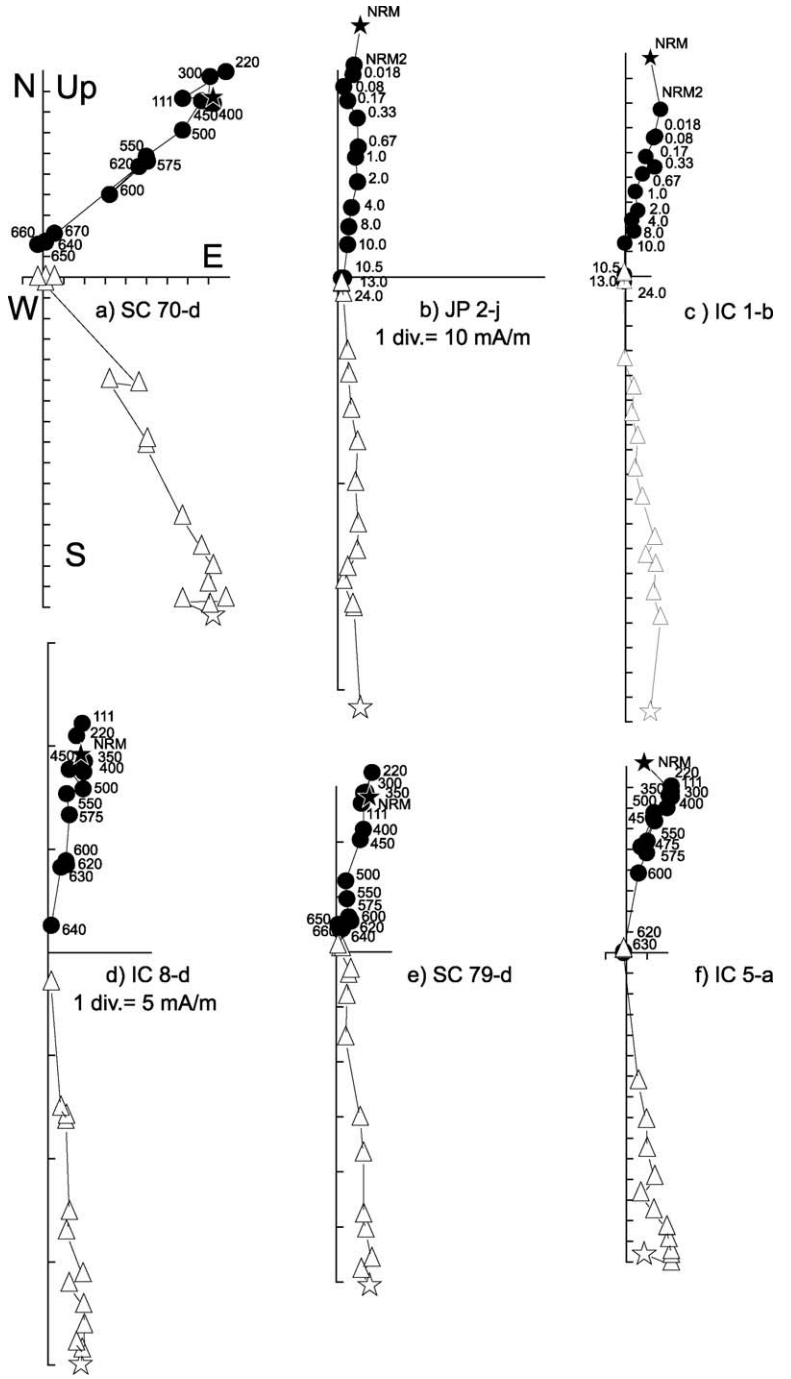


Fig. 3. (a–f) Representative examples of progressive thermal and chemical demagnetization behavior of UMG samples using orthogonal demagnetization diagrams (Zijderveld, 1967), which contain the north to northeast-directed, positive inclination, normal polarity magnetization. In each diagram, the endpoint of the magnetization vector is projected onto the horizontal plane (solid circles) and the vertical plane (open triangles) using, in all cases, the *E–W* axis as the common projection axis. Demagnetization steps are indicated along data points on the horizontal projection in degrees C or, for chemical demagnetization, as the cumulative hours of leaching in reagent HCl. Unless indicated, the tic mark on both axes is 1.0 mA/m. NRM2 is the second NRM measurement, after cuts are made into the specimen to increase the surface area.

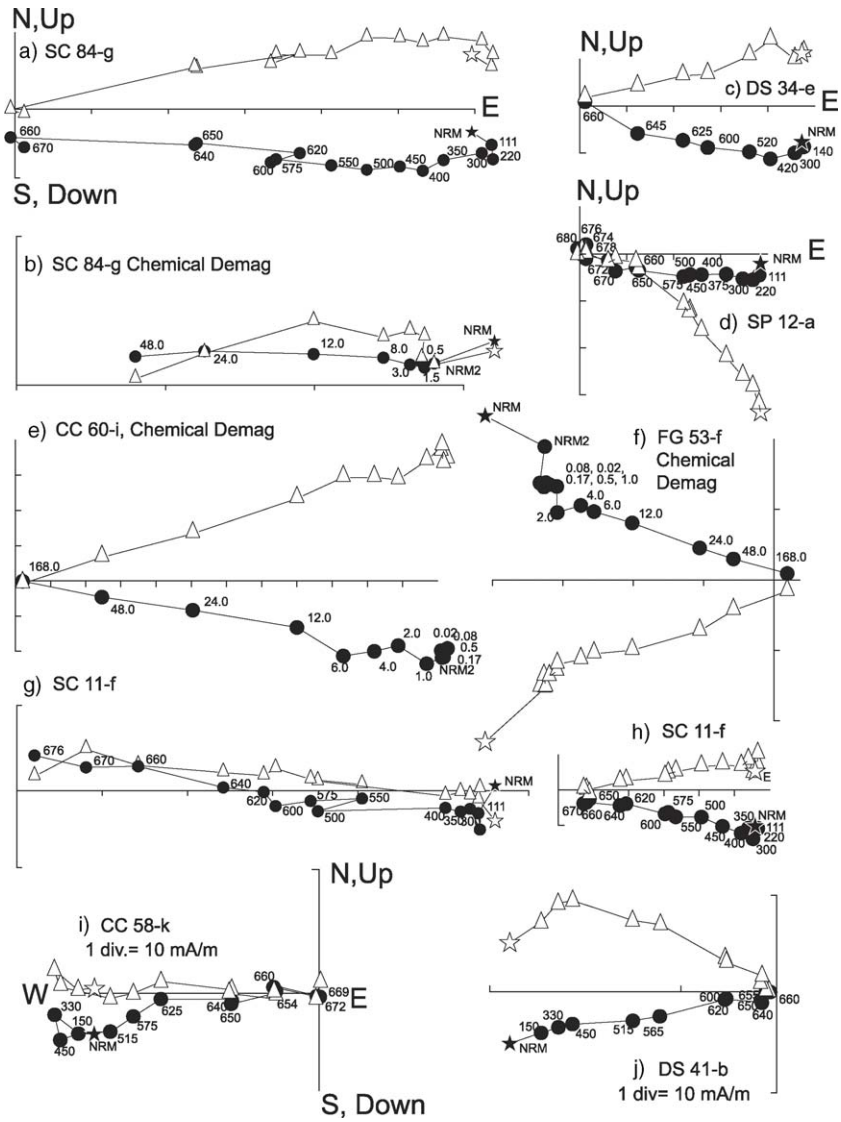


Fig. 4. (a–j) Representative examples of progressive thermal and chemical demagnetization behavior of UMG samples using orthogonal demagnetization diagrams (Zijderveld, 1967), which contain the east-directed (or antipode) shallow, positive to negative inclination magnetization. Conventions as in Fig. 3.

Specific details of the paleomagnetic results from UMG rocks are provided below by locality, in geographic order from west to east. Sites at each locality were collected in stratigraphic order, without duplication; however, correlation of each specific stratigraphic interval among localities is not well understood. A summary of paleomagnetic data is given in Tables 1 and 2.

4.1.1. Sheep Creek Canyon, north-central Uinta Mountains

Sheep Creek Canyon exposes part of the northern range-bounding Uinta reverse fault system (Fig. 2) and nearly 700 m of UMG and younger Phanerozoic strata (Schell, 1969; Sprinkel et al., 2000). About 3 km south of the fault zone, UMG strata strike east–west and have near-horizontal dips; however, their strike changes to west–northwest and steep north-tilted to overturned dips approaching the bounding fault system. A total of 45 sites (350 samples) were collected in UMG strata along about 3 km of the Sheep Creek drainage, from the overturned beds adjacent to the fault system to nearly horizontal beds away from the fault.

Of the 45 sites, 42 have site mean confidence estimate values (α_{95}) of less than 15° and were used to calculate

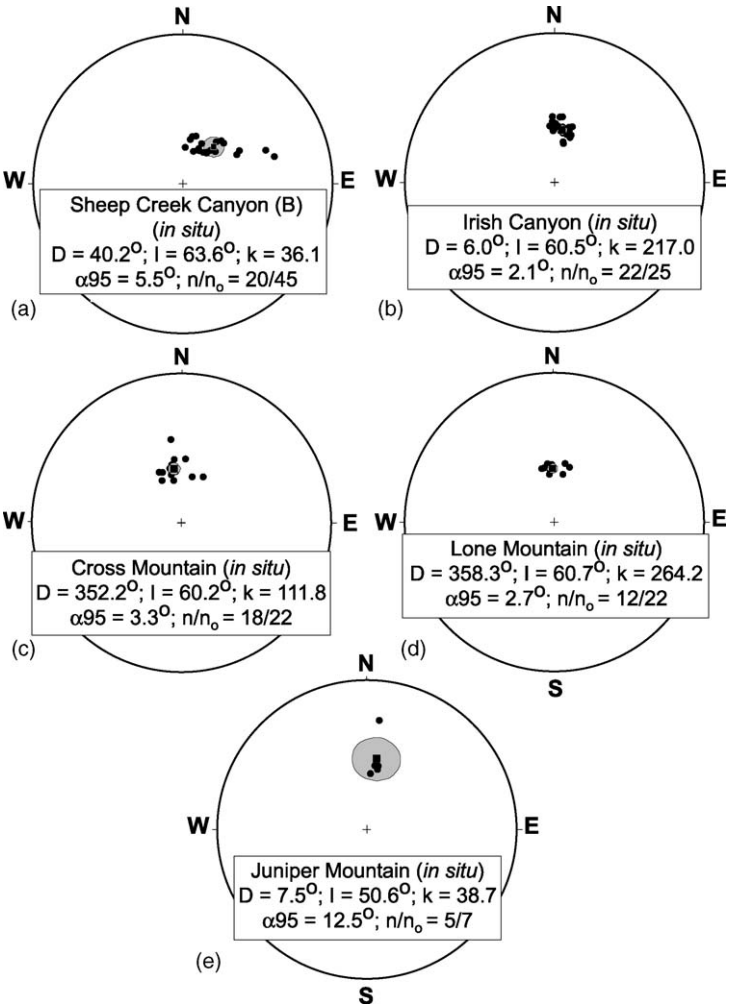


Fig. 5. (a–e) Equal area projections of paleomagnetic site mean data from the UMG localities studied, which contain the north to northeast-directed, positive inclination, normal polarity magnetization. Closed (open) symbols refer to lower (upper) hemisphere projections.

the locality mean direction. Twenty three sites carry an east declination and shallow, negative to positive inclination component, while the remaining 20 sites carry a moderate positive inclination and north to northeast-directed component.

A group mean determination for the 23 east-directed magnetization sites in stratigraphic coordinates is, Decl. = 96.2°, Incl. = −5.6°, α_{95} = 4.7°, k = 42.9, pole of 6.5°N, 158.4°E (Fig. 6a and Table 3). The north-directed and steep, positive inclination magnetization from 20 sites, provided a mean of, in *in situ* coordinates, Decl. = 40.2°, Incl. = 63.6°, α_{95} = 5.5°, pole at 62.2°N, 43.7°W, and in stratigraphic coordinates of Decl. = 43.4°, Incl. = 4.7°, α_{95} = 10.1°.

The change in dip from essentially flat-lying to overturned across the south to north traverse at this locality, facilitated tilt tests using both magnetizations. For the east and shallow magnetization, comparing the *in situ* and tilt-corrected kappa values for the mean of 23 sites ($k_{\text{in situ}}$ = 38.01, $k_{\text{corrected}}$ = 42.9; $k_{\text{corrected}}/k_{\text{in situ}}$ = 1.13) and the appropriate F-distribution value for the 10% significance level (N/N_0 = 23/45, $F_{44,\,44}$ = 1.42) indicates that the grouping after untilting is not significant at the 90% level ($k_{\text{corrected}}/k_{\text{in situ}} < F_{44,\,44}$). Unfortunately all of the bedding strikes are roughly east–west, and so no matter how tilted the strata are, progressive restoration of the beds to horizontal does very little to rotate the east–west oriented magnetization vectors during unfolding. Comparing north-directed magnetization *in situ* and unfolded kappa-values ($k_{\text{in situ}}$ = 36.07, $k_{\text{corrected}}$ = 11.33; $k_{\text{corrected}}/k_{\text{in situ}}$ = 0.31) with the F-distribution value for the 10% significance level (N/N_0 = 20/45, $F_{38,\,38}$ = 1.52)

Table 1
Paleomagnetic site information and statistical parameters for the UMG east (west)-directed component

Site number	Strike	Dip	N/N_0	*In Situ Site Mean*				*Structurally corrected site means*	
				Decl.	Incl.	α_{95}	k	Decl.	Incl.
Sheep Creek Canyon									
SC 2[a]	325	81	5/11	104.5	36.0	11.5	45.0	97.9	−25.3
SC 5[a]	324	80	4/12	102.6	58.7	12.7	53.0	77.4	−11
SC 6[a]	325	78	7/12	114.0	31.3	12.6	23.8	105.7	−18.8
SC 9[b]	293	43	3/20	97.5	−8.2	17.1	52.9	107.2	−16.5
SC 10[a]	301	55	3/3	81.2	33.3	11.9	108.4	71.3	−7.1
SC 11	296	39	12/12	98.1	−12.5	4.1	115.0	110	−20.9
SC 12	297	36	12/14	95.3	6.5	5.3	69.2	95.5	−7.1
SC 13	286	38	2/7	88.7	15.6	6.5	1502.0	83	2
SC 13[a]	286	38	3/7	82.3	32.9	12.5	98.1	68	12.7
SC 14	301	39	4/15	105.3	6.4	3.4	718.1	104.7	−4.7
SC 15	298	30	4/4	108.0	6.4	11.6	63.8	106.2	0.6
SC 16	315	41	5/5	105.9	6.0	9.5	65.6	108.5	−13.8
SC 17	308	41	9/9	90.5	10.2	7.5	48.1	91.9	−15
SC 19	303	40	5/14	96.2	11.7	9.6	64.6	94.8	−7.4
SC 20	304	35	11/11	97.4	6.0	5.7	64.6	98.5	−9.8
SC 21[b]	305	52	3/13	273.1	62.9	20.7	36.5	0	47.5
SC 22	287	30	14/17	90.8	22.3	6.4	39.3	82.1	11.5
SC 23	287	30	8/12	98.3	0.5	7.5	56.0	99.2	−3.9
SC 24	267	20	4/5	95.1	−1.2	3.6	665.2	95	1.6
SC 26	257	22	8/8	89.8	−0.4	5.9	55.9	89	4.4
SC 71	294	33	5/8	99.5	−1.8	11.3	47.2	102.7	−9.4
SC 72[b]	318	25	5/8	96.4	18.4	15.9	24.0	93.2	1.1
SC 75	294	39	7/11	89.6	0.1	13.2	22.0	94.5	−15
SC 77	284	32	8/8	90.6	11.1	7.5	66.3	86.8	2.5
SC 80	296	27	7/7	86.6	20.7	10.2	36.0	81	6.1
SC 83	284	30	8/8	103.3	−11.1	2.5	502.2	109	−9.9
SC 84	278	95	7/7	90.9	−0.9	7.3	70.1	99.5	−7
SC 85[a]	280	30	8/8	77.2	17.9	8.2	46.4	71.7	4.7
SC 86	267	23	7/7	88.5	9.9	8.3	53.5	84.5	9.7
SC 87	279	35	7/7	89.6	−11.8	7.3	69.4	98.2	−15
SC 88[a]	292	27	5/6	74.3	10.7	11.6	44.3	73.4	−6.2
SC 89	247	15	9/12	99.3	−19.5	11.0	23.0	102.7	−11.1
SC 90	247	15	6/6	91.1	−21.8	9.6	50.1	95.6	−15.1
Browne Lake									
BL 54[c]	278	5	–	–	–	–	–	–	–
BL 55[a]	278	5	5/5	41.7	57.9	6.4	114.9	37.8	53.7
BL 56	277	11	5/10	82.9	−5.8	13.1	35.1	84.3	−8.3
Carter Creek									
CC 57	285	12	8/11	263.2	−13.4	5.8	–	261	−8.7
CC 58	232	10	10/12	266.4	−6.7	4.4	–	265	−12.3
CC 59	277	9	5/8	270.4	−11.8	12.9	–	268.6	−10.6
CC 60	300	5	4/8	272.0	−5.0	3.4	–	271.7	−2.6
CC 61	281	1	9/11	263.9	−5.9	11.7	–	263.8	−5.6
Dowd Springs non-fold test rocks									
DS 29[b]	–	0	7/10	98.7	−20.1	30.7	4.8	98.7	−20.1
DS 29[a]	–	0	5/10	281.1	−42.2	14.3	30.3	281.1	−42.2
DS 30[a]	107	4	6/10	290.9	−72.2	8.1	69.2	302.9	−71.5
DS 31[b]	217	3	8/11	107.1	3.3	20.6	8.2	107	6.1
DS 32[b]	–	0	5/10	282.0	−36.1	24.0	11.1	282	−36.1
DS 33[b]	132	7	3/10	97.0	−14.7	19.9	39.5	95.7	−10.6
DS 34	132	7	9/10	101.9	−5.0	10.8	23.7	101.6	−1.5
DS 35	99	4	10/11	103.8	−7.2	8.8	31.1	103.3	−7.5

Table 1 (*Continued*)

Site number	Strike	Dip	N/N_0	*In Situ Site Mean*				*Structurally corrected site means*	
				Decl.	Incl.	α_{95}	k	Decl.	Incl.
DS 36	102	6	9/11	108.7	−23.0	9.3	31.4	106.1	−23.6
DS 37	120	10	9/10	107.6	6.6	12.4	18.3	108.9	8.6
DS 38	102	12	15/15	91.7	−12.7	10.7	13.7	89.3	−10.3
Dowd Springs fold test rocks									
DS 39	63	19	4/5	261.7	−0.2	11.5	48.33	260.8	11.5
DS 40	63	19	4/5	257.8	−38.7	12.7	64.10	270.5	12.7
DS 41	63	19	3/5	261.2	−19.4	8.5	139.59	266.3	8.5
DS 42	221	14	3/4	271.2	−0.6	14.8	46.69	270.3	14.8
DS 43	221	14	3/5	276.4	8.2	12.2	32.10	276.7	12.2
DS 44	221	14	5/5	271.4	−6.8	7.9	103.90	269.4	7.9
Flaming Gorge Section									
FG-AW-01	236	18	5/6	270.8	−9.3	16.5	57.0	267	−19
FG-AW-02	236	18	6/6	264.4	−10.5	9.4	51.4	260	−18
FG-AW-03	236	18	4/5	274.0	−7.6	11.8	110.2	271	−19
FG-AW-04	236	18	6/6	255.6	4.6	11.4	35.4	257	−1
FG-GA-01	232	15	5/5	264.9	−5.3	6.6	135.2	261	−18
FG-GA-02	220	20	6/6	268.4	−1.0	20.1	15.5	266	−16
FG-GA-03	229	17	5/6	266.5	−3.4	13.3	34.0	265	−13
FG-GA-04	205	17	5/6	262.2	7.8	10.4	54.7	262	−6
FG-GA-05[c]	236	16	–	–	–	–	–	–	–
FG-GA-06	238	18	5/6	275.1	0.9	9.1	72.0	274	−10
FG-GA-07	237	18	6/6	271.1	−3.1	8.3	65.7	269	−13
FG-GA-08	237	18	6/6	259.1	−3.0	6.7	101.8	257	−10
FG 48	250	18	4/11	275.2	−8.2	4.4	–	271.7	−15.4
FG 49	273	18	6/10	267.7	−5.7	8.1	–	266.2	−3.8
FG 50[a]	254	20	7/11	295.3	−10.3	8.9	–	290.6	−23
FG 51	268	16	5/7	275.1	−3.1	3.2	–	274	−4.9
FG 52	268	18	5/13	279.2	−12.1	8.3	–	274.9	−15
FG 53	296	15	4/5	275.5	2.5	7.5	–	276.8	7.6
Bull Canyon									
BC 01	29	13	8/8	270.6	5.9	6.0	101.1	267.9	8.8
BC 02[b]	55	16	8/8	280.2	6.2	16.6	13.9	277.2	7.3
BC 03	16	19	6/10	267.3	−5.9	14.4	23.4	267	12.1
BC 04[b]	51	20	9/9	276.4	−2.2	20.5	8.0	271.8	1.2
BC 05	51	20	7/9	274.2	10.0	14.2	22.5	272.3	8.7
BC 06	51	20	10/10	269.0	5.1	5.8	71.9	264.9	12.1
BC 07	49	12	9/9	279.4	2.5	8.4	44.4	277.9	5.8
BC 08	89	17	9/14	282.9	2.3	9.2	37.7	281.8	6.2
BC 09[a]	19	12	10/10	2.1	56.7	4.4	121.3	20.1	58.2
BC 10[b]	99	2	7/13	267.1	8.2	17.9	14.5	269	10.5
Home Montain Canyon									
UMG-GA-01	112	82	7/8	289.5	21.2	15.9	24.1	271	0
UMG-GA-02	116	52	7/8	284.3	34.7	14.1	19.2	261	12
UMG-GA-03	113	66	5/6	287.9	31.7	18.4	18.2	262	8
Sparks–Talamantes Creek									
SP 01	306	58	9/9	105.2	22.1	6.5	63.5	96.3	−4.6
SP 02[a]	295	62	5/7	67.8	59.2	14.3	29.7	45.4	4.1
SP 03	313	70	5/6	92.1	30.5	11.9	42.3	87.2	−20.9
SP 04	299	61	7/7	95.9	35.6	9.2	43.9	77.4	0.2
SP 05	315	63	8/8	88.8	34.6	11.6	23.7	81.3	−15.8
SP 06	324	66	7/7	99.9	26.0	8.6	50.7	98.6	−23.1
SP 07	299	61	4/8	104.2	31.6	14.0	43.9	84.6	3.7
SP 08	312	65	7/8	95.7	20.0	9.2	43.8	96.3	−21.1

Table 1 (*Continued*)

Site number	Strike	Dip	N/N_0	*In Situ Site Mean*				*Structurally corrected site means*	
				Decl.	Incl.	α_{95}	k	Decl.	Incl.
SP 09[b]	314	71	3/7	94.9	28.4	28.42	20.2	91.3	−21.7
SP 10	313	56	6/8	103.1	36.3	10.2	44.5	87.3	−0.1
SP 11[b]	301	59	6/8	101.7	21.3	15.1	20.7	92.9	−4.4
SP 12	313	59	8/8	95.6	35.9	7.0	63.8	83.4	−6.9

N/N_0 is ratio of samples used in the calculation of the site mean direction to the total number of samples collected and measured; strike/dip, dip direction is 90° clockwise from strike; Decl. and Incl. are declination and inclination of the site mean directions; α_{95} is semi-angle of the cone of confidence about the site mean direction; k is precision parameter (Fisher, 1953).

[a] rejected because direction greater than two angular standard deviations from the estimated mean.

[b] rejected due to $\alpha_{95} > 15.0°$.

[c] Designates those sites that had no interpretable results.

showed that the increase in dispersion of site means after untilting is significant at the 90% level.

4.1.2. Browne Lake and Carter Creek, north-central Uinta Mountains

Four sites (total of 72 samples) were taken from Browne Lake, and five sites (total of 55 samples) were taken from Carter Creek (Fig. 2). UMG rocks sampled at Browne Lake are about 250 m stratigraphically above those taken from the sites at Sheep Creek Canyon. However, only one of the four sites was accepted, and provided a site mean of, in stratigraphic coordinates, Decl. = 84.3°, Incl. = −8.3°, $\alpha_{95} = 13.1°$, $N/N_0 = 5/10$, $k = 35.12$, VGP of 1.6°S, 167.4°E (Fig. 6b and Table 3). The sites at Carter Creek are about 50 m stratigraphically below those collected at Browne Lake. All five sites, with some samples showing an east-directed and shallow magnetization, were accepted for a locality group mean of, in stratigraphic coordinates, Decl. = 266.0°, Incl. = −8.0°, $\alpha_{95} = 5.4°$, $N/N_0 = 5/5$, $k = 225.6$, pole of 5.7°S, 160.1°E (Fig. 6c and Table 3).

4.1.3. Dowd Springs, north-central Uinta Mountains

Sixteen sites (total of 137 samples) were collected from UMG rocks at the Dowd Springs sampling locality (Fig. 2), six of which (29 of the 137 total samples) were established in both limbs of an upright, open anticline (Trend/Plunge about 260/15). Five of the 10 sites taken from very gently tilted UMG strata yield accepted mean directions, two of which are based on dual polarity magnetizations, and provide a group mean magnetization of, in stratigraphic coordinates, Decl. = 101.8°, Incl. = −6.9°, $\alpha_{95} = 12.1°$, $N/N_0 = 5/10$, $k = 40.9$, pole of 11.2°N, 155.6°E (Fig. 6d and Table 3).

A fold test of the six sites collected from the upright fold yields a decrease in site mean dispersion with a simple tilt-correction ($k_{\text{in situ}} = 19.75$, $k_{\text{corrected}} = 34.9$; $k_{\text{corrected}}/k_{\text{in situ}} = 1.77$); however, the improved site mean grouping after unfolding is not significant at the 90% level ($N/N_0 = 6/6$, $F_{10,10} = 2.32$). Like the Sheep Creek Canyon fold, all of the bedding orientations from Dowd Springs have roughly the same orientation as the magnetization vectors. The group mean for the six sites is, in stratigraphic coordinates, Decl. = 269.0°, Incl. = −11.9, $\alpha_{95} = 11.5°$, $N/N_0 = 6/6$, $k = 34.9$, pole of 4.7°S 156.7°E (Fig. 6d and Table 3).

4.1.4. Flaming Gorge dam, north-central Uinta Mountains

Eighteen sites (127 samples total) were collected in the Flaming Gorge reservoir dam area, with six sites collected about 2 km west of the dam and 12 sites collected along the boat landing road immediately below the dam (Fig. 2). Most specimens yielded west-directed magnetizations resolved over a range of high laboratory unblocking temperatures and after chemical treatment beyond about 20 h (Fig. 4f). A group mean, based on 16 of the 18 sites, is in stratigraphic coordinates, Decl. = 267.1°, Incl. = −9.8°, $\alpha_{95} = 4.5°$, $k = 66.8$, and pole at 5.4°S, 158.9°E (Fig. 6e and Table 3).

4.1.5. Irish Canyon, Browns Park, eastern Uinta Mountains

Irish Canyon (Fig. 2) is a 3-km long canyon that, in sections, has incised up to 200 m through a north–northwest to northwest striking, northeast dipping (∼30°) section of UMG and younger Phanerozoic strata at the northeast edge of Browns Park valley. The trace of the main northern Uinta bounding fault lies north and east of Irish Canyon and is oriented NNW.

Twenty-five sites were collected across about 70 m of section near the western mouth of the canyon (a total of 178 samples). Samples from 22 of these sites provided an *in situ* group mean of, Decl. = 6.0°, Incl. = 60.5°,

Table 2
Paleomagnetic site information and statistical parameters for the UMG north-directed component

Site number	Strike	Dip	N/N_0	*In Situ Site Mean*				Structurally *corrected site means*	
				Decl.	Incl.	α_{95}	k	Decl.	Incl.
Sheep Creek Canyon									
SC 1	321	80	10/11	33.4	69.6	3.0	259.7	44.9	-9.5
SC 2	325	81	6/11	9.7	65.5	4.2	262.2	37.7	−8.4
SC 3	322	82	11/12	24.6	69.0	3.6	161.4	42.3	−10.7
SC 4	323	81	11/11	37.6	68.9	2.8	262.0	47.4	−11.4
SC 5	324	80	8/12	37.5	68.3	3.5	255.4	47.9	−10.8
SC 6	325	78	4/12	26.6	69.9	5.4	296.3	45.5	−5.8
SC 7	325	81	12/12	25.1	65.8	3.6	146.0	42.9	−12
SC 8	321	85	10/11	45.5	65.0	6.2	61.4	48.5	−19.9
SC 9	293	43	17/20	38.7	60.7	5.4	45.2	31	18.5
SC 14	301	39	10/15	61.8	56.3	6.7	53.3	48.6	20.3
SC 19	303	40	9/14	73.5	36.3	8.0	42.7	64.6	3.4
SC 21	299	43	10/13	68.0	40.0	7.1	47.1	57.9	3.7
SC 46	352	102	6/6	4.1	70.2	2.5	718.8	61.9	−15.4
SC 47	354	102	6/6	18.2	71.4	4.5	221.5	66.1	−19
SC 70	302	44	9/9	45.4	58.4	7.0	55.6	39.2	15
SC 74	294	39	5/8	39.7	66.4	9.5	65.8	31	28
SC 78	302	20	8/8	15.8	63.0	2.0	739.8	21.9	43.5
SC 79	262	23	8/8	11.5	63.7	3.8	218.5	3.4	41.5
SC 81	282	34	6/7	59.7	53.9	9.1	55.1	41.2	26.6
SC 82	291	41	8/8	41.8	58.3	8.0	49.1	32.4	18.7
Irish Canyon Section									
IC-AW-01	342	26	4/6	354.8	58.6	6.1	414.6	25	46
IC-AW-02	330	/6	6/6	357.6	61.6	7.6	79.0	25	44
IC-AW-03	340	28	5/6	357.1	53.7	9.4	67.1	24	39
IC-AW-04	340	28	6/6	356.8	59.0	2.5	741.5	28	44
IC-AW-05	330	29	6/6	358.1	62.0	3.7	420.1	26	43
IC-AW-06	332	29	6/6	359.1	60.8	1.9	1287.0	25	42
IC-AW-07[c]	332	29	–						
IC-AW-08	332	29	6/6	1.8	60.4	3.4	399.7	28	39
IC-AW-09	332	29	5/6	357.8	56.6	2.4	1047.9	24	39
IC-AW-10	328	26	6/6	0.4	58.8	5.0	231.1	22	41
IC-AW-11	328	26	5/6	0.8	59.7	2.5	1302.2	23	40
IC-AW-12	325	38	4/6	5.5	53.9	12.3	101.6	26	33
IC-AW-13[c]	322	26	–						
IC-AW-14[c]	322	26	–						
IC 01	6	19	8/9	8.5	59.3	5.1	119.2	36.5	53.7
IC 02	6	19	7/8	13.0	68.3	2.6	553.6	49.2	59.8
IC 03	6	19	9/10	12.1	67.4	9.3	31.4	47.6	59.3
IC 05	347	26	8/9	9.9	62.7	2.8	294.1	39.5	46.1
IC 06	334	19	8/8	16.2	60.8	1.3	1702.2	32.7	45.9
IC 07	309	23	7/8	19.4	62.6	3.0	415.3	27.3	40.4
IC 08	317	28	8/8	19.1	64.2	2.9	527.6	32	37.9
IC 10	341	21	8/9	17.9	57.9	2.1	722.6	35.8	42.6
IC 11	329	33	9/9	14.7	58.8	3.4	232.1	34	31
IC 12	349	27	8/8	4.7	53.9	4.6	231.0	30.9	40.4
IC 14	340	25	7/8	14.0	63.0	1.8	1103.5	38.2	44.4
Cross Mountain									
CR 1	30	15	7/8	350.1	60.7	8.6	43.1	18.3	67.5
CR 2	30	15	10/12	350.4	57.9	3.1	226.4	15.2	64.9
CR 3	30	15	12/12	349.0	61.6	6.5	42.5	18.4	68.5
CR 4	32	8	9/10	25.3	62.2	12.2	16.7	40.5	62.1
CR 5	115	18	11/11	339.4	60.6	7.6	33.8	304.3	69.2
CR 6	115	18	12/13	349.6	66.6	9.2	21.2	301	76.6
CR 7	114	18	9/11	335.3	64.7	8.4	34.6	292	71.3

Table 2 (*Continued*)

Site number	Strike	Dip	N/N_0	*In Situ Site Mean*				Structurally *corrected site means*	
				Decl.	Incl.	α_{95}	k	Decl.	Incl.
CR 8	157	33	10/12	348.0	63.3	7.1	42.8	293.8	52.7
CR 9	54	19	10/10	138.4	45.8	46.6	1.8	139.6	26.9
CR 10[b]	54	19	10/12	19.2	71.6	16.1	9.0	82.1	72.9
CR 11	54	19	9/9	349.3	58.9	3.8	163.1	17.6	74.1
CR 12	57	14	11/11	350.2	59.6	4.3	103.1	6.4	71.7
CR 13	57	14	5/10	352.8	43.3	5.3	170.0	1	55.5
CR 14	199	11	12/12	349.9	60.2	3.9	112.6	336.1	53.7
CR 15	24	10	8/8	353.6	61.0	2.0	646.4	11.9	64.7
CR 16	24	10	7/10	351.9	60.7	2.7	446.6	9.8	64.7
CR 17	319	15	6/9	13.1	64.3	9.4	43.5	25.1	51.2
CR 18	92	8	10/10	336.2	59.8	5.5	70.3	328.3	66.8
CR 20	37	13	8/8	353.6	54.9	4.0	171.6	11.3	62.4
CR 21	22	12	8/12	3.6	54.9	5.3	97.1	21.2	56.9
CR 22[b]	37	14	5/10	339.3	44.9	17.1	16.7	349.7	56.1
CR 22[b]	37	14	5/10	104.9	−51.2	18.9	13.9	94.8	−63.7
Lone Mountain									
LN 1	317	27	7/10	353.4	57.8	6.0	86.8	14.2	37.6
LN 2[b]	317	27	10/10	19.0	46.7	19.3	6.5	26.7	21.9
LN 4	317	27	10/10	15.6	58.9	13.4	12.6	28	34.2
LN 8	310	16	11/11	355.8	61.9	3.6	145.9	10	49
LN 9	340	30	10/10	10.6	57.3	5.6	67.2	34.8	36.2
LN 10	287	31	9/9	346.9	60.0	3.5	197.7	359.9	31.3
LN 11	287	31	9/9	349.1	59.9	4.1	139.3	1.1	30.9
LN 12	287	31	8/9	357.1	61.7	10.3	26.2	6.1	31.7
LN 13	274	30	10/10	352.5	59.8	2.5	327.8	357.3	30.1
LN 14	283	45	8/11	357.5	58.4	4.1	164.5	4.7	14.2
LN 15	285	22	6/8	354.3	63.9	6.4	92.9	2.8	42.7
LN 17	304	30	6/12	10.4	63.6	7.8	62.5	21.5	34.9
LN 17[a]	304	30	6/12	195.8	67.0	4.6	178.4	75.3	79.3
LN 20	316	24	8/8	355.6	61.8	5.4	92.9	16.1	43
Juniper Mountain									
JP 1[b]	134	9	10/10	0.8	64.7	22.3	4.9	343.6	70.3
JP 2	134	9	8/9	10.1	54.5	6.5	64.5	1.1	61.6
JP 3	139	12	8/9	10.2	56.5	4.7	108.1	354.6	64.8
JP 4	139	12	9/9	7.2	54.7	9.9	24.8	352.2	62.6
JP 5	139	12	10/10	3.8	59.3	6.2	54.9	344.5	66.3
JP 6	142	11	9/9	6.5	27.5	14.5	12.1	1.5	34.9
JP 7[a]	142	11	5/8	272.0	48.9	10.5	43.0	265.5	40.1

N/N_0 is ratio of samples used in the calculation of the site mean direction to the total number of samples collected and measured; strike/dip, dip direction is 90° clockwise from strike; Decl. and Incl. are declination and inclination of the site mean directions; α_{95} is semi-angle of the cone of confidence about the site mean direction; k is precision parameter (Fisher, 1953).

[a] rejected because direction greater than two angular standard deviations from the estimated mean.

[b] rejected due to $\alpha_{95} > 15.0°$.

[c] Designates those sites that had no interpretable results.

$\alpha_{95} = 2.1°$, $k = 217.0$, pole of 85.4°N, 146.5°E, and in stratigraphic coordinates, Decl. = 35.9°, Incl. = 45.7°, $\alpha_{95} = 5.6°$, pole of 79.7°N, 37.8°W (Fig. 5b).

4.1.6. Bull Canyon, Browns Park, eastern Uinta Mountains

Ten stratigraphically ordered sites (100 samples total) were taken from north to northeast striking, southeast dipping (∼20°) UMG strata along Bull Canyon (Fig. 2). Sampling began about 600 m stratigraphically below the locality at Irish Canyon and covered about 100 m of section.

All but two of the ten sites investigated required remagnetization circle analysis (McFadden and McElhinny, 1988) to analyze the thermal demagnetization data (Fig. 7a–d). The other two sites (BC3

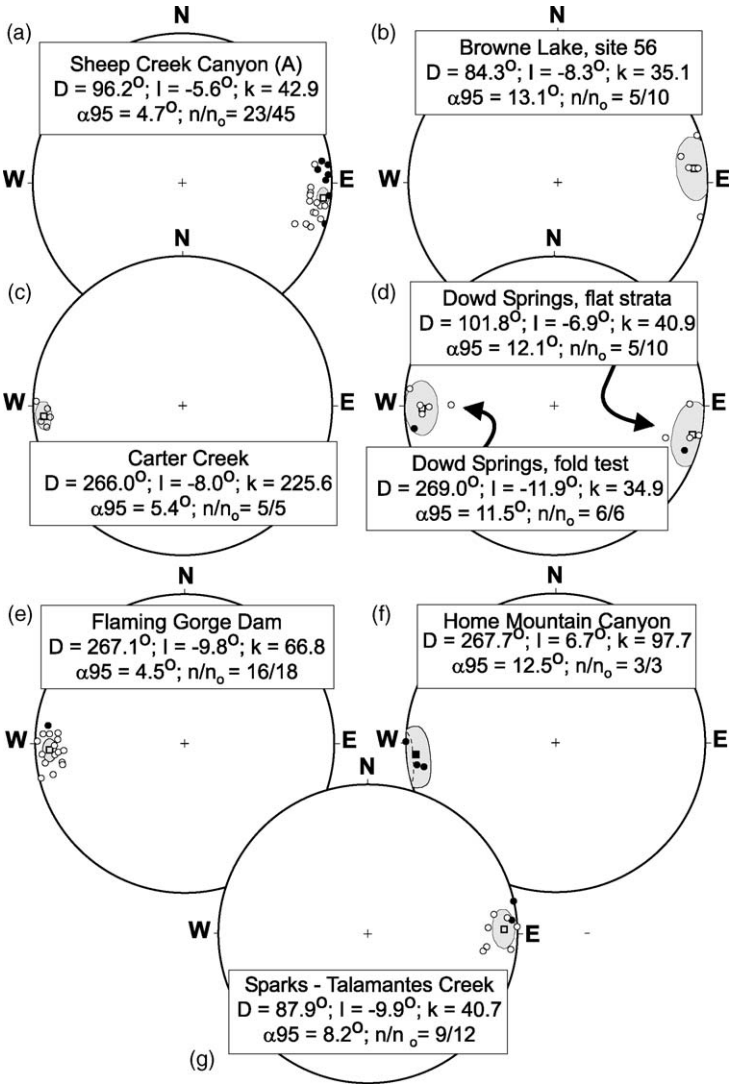


Fig. 6. (a–g) Equal area projections of paleomagnetic site mean data from the UMG localities studied, which contain the east-directed (or antipode) shallow, positive to negative inclination magnetization. All projections are in stratigraphic coordinates, with structural corrections applied about the present strike axis at each site as described in text. Closed (open) symbols refer to lower (upper) hemisphere projections.

and BC9), yield well-defined stable endpoint behavior. Site BC9 yields a north-directed, positive inclination mean of, in stratigraphic coordinates, Decl. = 20.1°, Incl. = 58.2°, α_{95} = 4.4°, which is similar to some of the magnetizations isolated at the Sheep Creek locality, and provides a pole of 74.5°N, 161.5°E. Site, BC3, yields a west-directed and shallow inclination magnetization of, in stratigraphic coordinates, Decl. = 267.0°, Incl. = 12.1°, α_{95} = 14.4°, VGP of 1.7°N, 167.8°E. All samples from the other eight Bull Canyon sites exhibit trajectories showing the removal of steep, positive inclination, west to north-directed magnetizations at low laboratory unblocking temperatures and, at higher temperatures, the partial isolation of a west-directed, shallow magnetization (Fig. 7). The mean direction from site BC3 was taken to represent a set point magnetization for the great circle trajectories defined by samples from the other eight sites. Site means determined using this technique were combined with BC 3 for a group mean of, in stratigraphic coordinates, Decl. = 272.0°, Incl. = 9.0°, α_{95} = 5.9°, pole of 4.5°N, 163.3°E (Fig. 7g and Table 3).

4.1.7. Home Mountain Canyon, eastern Uinta Mountains

Three sites (22 samples total) were established in basal UMG rocks exposed along Home Mountain Canyon (Fig. 2), just west of the Utah, Colorado boarder.

Table 3
Paleomagnetic Pole Information for the Uinta Mountain Group

Pole location	N/N_0	Decl.	Incl.	α_{95}	GeoLong.	GeoLat.	PLong.	PLat.
Sheep Creek	23/51	96.2	−5.6	4.7	250.3	40.9	158.4	6.5
Browne Lake	1/3	84.3	−8.3	13.1	250.5	40.5	167.4	−1.6
Carter Creek	5/5	266.0	−8.0	5.4	250.6	40.5	160.1	−5.7
Dowd Springs, flat	5/10	101.8	−6.9	12.1	250.6	40.5	155.6	11.2
Dowd Springs, fold	6/6	269.0	−11.9	11.5	250.6	40.5	156.7	−4.7
Flaming Gorge	16/18	267.1	−9.8	4.5	250.7	40.6	158.9	−5.4
Bull Canyon	10/10	272.0	9.0	5.9	251.2	40.9	163.3	4.5
Home Mountain Canyon	3/3	267.7	6.7	12.5	250.8	41.0	164.8	0.5
Sparks–Talamantes Creek	9/12	87.9	−9.9	8.2	251.2	40.9	166.3	1.7

N/N_0 is ratio of sites used in the calculation of the paleomagnetic pole to the total number of sites collected; Decl and Incl are declination and inclination of the mean paleomagnetic directions for the stated geographic area; α_{95} is semi-angle of the cone of confidence about the mean direction; GeoLong. and GeoLat. are the latitude and longitude of the sampling area location; Plong. and Plat, are the latitude and longitude of the mean paleomagnetic pole; the value of the paleomagnetic pole in the table is based on the tilt-corrected site mean directions.

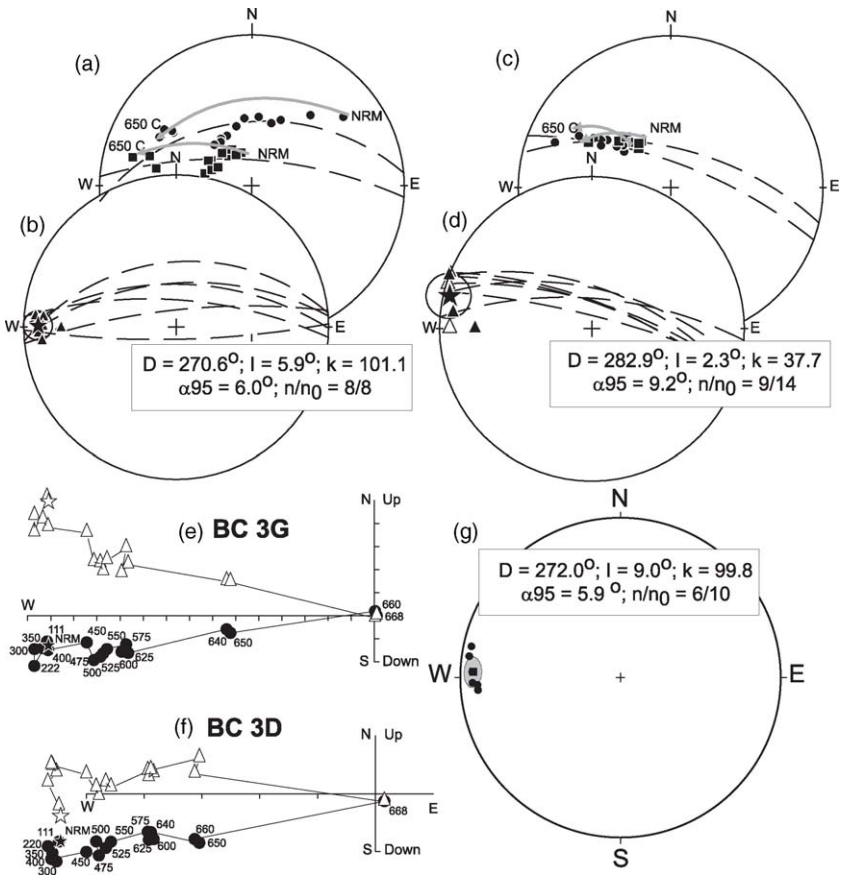


Fig. 7. Paleomagnetic data from the Bull Canyon locality. (a and b) equal area projections showing demagnetization trajectories and site mean determination for site BC1. (c and d) equal area projections showing demagnetization trajectories and site mean determination for site BC8. (e and f) Examples of progressive thermal demagnetization behavior for two samples from site BC3, which did not require remagnetization circle analysis, using orthogonal demagnetization diagrams (Zijderveld, 1967). Conventions as in Fig. 3. (g) Summary of Bull Canyon locality data using the six best defined site means.

Most samples yielded a west-directed and shallow magnetization that provided a locality mean, based on data from three sites, of, in stratigraphic coordinates, Decl. = 267.7°, Incl. = 6.7°, α_{95} = 12.5°, k = 97.7, pole of 0.5°N, 164.8°E (Fig. 6f and Table 3).

4.1.8. Sparks–Talamantes Creek, eastern Uinta Mountains

Twelve sites (91 samples total) were established in UMG rocks exposed at Talamantes Creek (Fig. 2), west of Sparks, Colorado, about 15 km northwest of Irish Canyon. The rocks lie in the footwall of a normal fault less than 1 km south of the main Uinta fault system. All samples yielded an east-directed and shallow magnetization that provided a well-grouped locality mean, based on data from nine sites, of, in stratigraphic coordinates, Decl. = 87.9°, Incl. = −9.9°, α_{95} = 8.2°, k = 40.7, pole of 1.7°N, 166.3°E (Fig. 6h and Table 3).

4.1.9. Cross Mountain, southeastern Uinta Mountains

Two sampling localities were established along the east margin of Cross Mountain, a 15 km long, 5 km wide anticline southeast of Browns Park, with 10 sites collected at one and 12 at another about 3 km to the south, for a total of 22 sites (231 samples).

A north-directed and steep positive inclination magnetization is pervasive in all samples from all sites at Cross Mountain and none possessed any evidence of the east or west and shallow magnetization identified at other localities. The north-directed positive inclination magnetization persisted to the highest laboratory unblocking temperatures, revealing no underlying magnetization. Chemical demagnetization also demonstrated that this remanence was the only component to the NRM. Given this behavior, we assumed that NRM directions of the remaining sites can be taken to approximate the ChRM. The combined grand mean of all NRM directions and the results of the completely demagnetized sites is, in *in situ* coordinates, Decl. = 352.2°, Incl. = 60.2°, α_{95} = 3.3°, k = 111.8, pole of 83.9°N, 177.8°E (Fig. 5c), and in stratigraphic coordinates, Decl. = 357.4°, Incl. = 67.8°, α_{95} = 6.9°.

4.1.10. Lone Mountain, southeastern Uinta Mountains

Lone Mountain (Fig. 2) is an isolated outcrop of UMG rocks directly northwest of Cross Mountain. Similar to Cross Mountain, all of the 22 sites collected yield NRM directions that are north-directed and of steep positive inclination; based on the behavior of results from Cross Mountain, only limited demagnetization was attempted (14 sites were completely demagnetized). A group mean of all NRM directions determined is, in *in situ* coordinates, Decl. = 358.3°, Incl. = 60.7°, α_{95} = 2.7°, k = 264.2, pole of 88.2°N, 156.4°W (Fig. 5d), and in stratigraphic coordinates, Decl. = 011.1°, Incl. = 35.2°, α_{95} = 7.0°.

4.1.11. Juniper Mountain, easternmost Uinta Mountains

Juniper Mountain is the easternmost exposure of the UMG, about 25 km from Cross Mountain. Seven sites were established in UMG rocks (64 samples total) and again, similar to results from Cross and Lone Mountain, NRM and limited thermal demagnetization results reveal the pervasive nature of a north-directed and steep, positive inclination magnetization. A grand mean of the NRM directions and those isolated in demagnetization provides, in *in situ* coordinates, Decl. = 007.5°, Incl. = 50.6°, α_{95} = 12.5°, k = 38.7 (Fig. 5e), pole of 88.2°N, 156.4°W, and in stratigraphic coordinates, Decl. = 356.0°, Incl. = 58.3°, α_{95} = 12.9°.

4.2. Rock magnetism and magnetic mineralogy, UMG Rocks

Thermal demagnetization of all UMG rocks investigated reveals that hematite is the primary carrier of what we interpret as an ancient, primary to near primary Neoproterozoic magnetization as well as the secondary magnetization that is pervasive at many UMG localities. High laboratory unblocking temperatures were observed for almost all of the samples, with most unblocking between about 600° and 670 °C. No substantial inflection is identified in normalized intensity decay plots between 500° and 580 °C, where a remanence carried by magnetite would be unblocked (Fig. 8). Alternating field (AF) demagnetization to peak fields of 120 mT on selected specimens from all UMG localities failed to randomize more than a few percent of the total NRM.

Rock magnetic experiments UMG rocks help to characterize the magnetic mineralogy. Single-axis IRM acquisition experiments with select UMG specimens yielded concave upwards curves similar to reference curves characteristic of hematite-dominated assemblages, with little to no contribution by magnetite (Dunlop and Ozdemir, 1997; Symons and Cioppa, 2000) (Fig. 9). Backfield DC demagnetization of IRM yielded coercivity of remanence values between 0.4 and 0.9 T (Fig. 9). Three-axis IRM thermal demagnetization experiments (Lowrie, 1990) on select UMG specimens from four different localities (Sheep Creek Canyon,

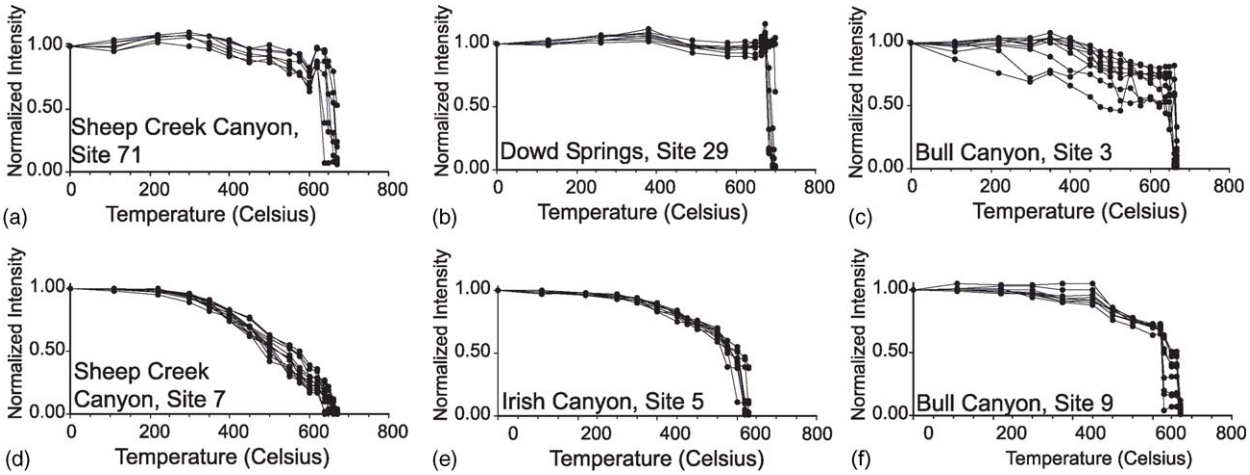


Fig. 8. (a–f) Normalized intensity vs. thermal demagnetization treatment plots for several UMG sites. For each site, the laboratory unblocking temperature for several independent samples is shown.

Bull Canyon, Irish Canyon and the Sparks–Talamantes Creek) (Fig. 10) are consistent with other results. The hard component (3.0 T acquisition) in these experiments consistently possessed the highest initial IRM intensity and the decay curve remained above those of the other components, being the last to completely unblock above 640 °C (Fig. 10). This response is characteristic of a hematite dominated-material (Dunlop, 1972; Lowrie, 1990). No goethite is observed by the hard component over the unblocking temperature range below about 120 °C (Dunlop and Ozdemir, 1997). All specimens examined had the intensity of the intermediate component (0.3 T) between the hard and soft (0.03 T) curves, and unblocking temperatures above 640 °C with no cusps in the thermal decay curve at lower temperatures (Fig. 10).

The soft (0.03 T) component of three-axis IRM thermal demagnetization is of negligible intensity and shows complete unblocking below about 580 °C. This result indicates a presence of a trivial amount of magnetite and/or maghemite, consistent with all other rock magnetic experiments. In three-axis IRM demagnetization, samples that carry the interpreted Neoproterozoic ChRM had, overall, higher initial IRM intensities of the hard (3.0 T) component than those from localities dominated by secondary magnetizations. However, NRM intensities show less of a correlation with the presence of a specific magnetization. In general, histograms of NRM intensities for specimens dominated by a secondary magnetization show median values that are up to an order of magnitude greater than those of specimens that contain only the east or west-directed and shallow magnetization.

Chemical demagnetization of NRM (Figs. 3b and c and 4b–f) shows that fine-grained, pigmentary hematite is the principal form of hematite present in all UMG rocks. The near-complete removal of the ChRM, in a cumulative time period of less than about 100 h for rocks from the central part of the Uintas, and typically by about 15 h for rocks from the eastern part (Fig. 11), demonstrates that most of the hematite carrying the magnetization in UMG rocks is fine-grained, pigmentary material (Collinson, 1966, 1974). The contribution to the overall remanence by detrital hematite appears to be minimal, as coarse, detrital hematite is not readily soluble in hydrochloric acid and would have been revealed by the presence of a still-measurable magnetization after essentially complete dissolution of pigment hematite.

Thin-section inspection of 11 samples from the Sheep Creek Canyon sampling locality revealed fine, translucent pigment hematite as the visibly dominant magnetic mineral. Pigmentary hematite coats grains and is intergrown with both calcite and secondary silica cement. Coarser (10 micron +) opaque mineral grains that are likely detrital hematite or magnetite were also observed, but at a very low frequency.

In general, the grain size of UMG rocks appears to correlate with NRM directions and the overall consistency of response to thermal demagnetization. Fine-grained (about 0.075–0.25 mm), well-indurated, silica-cemented samples tend to carry the ancient east or west declination, shallow inclination magnetization as a ChRM. This ChRM is typically well-defined in progressive thermal demagnetization and typically unblocks over a relatively discrete range of laboratory unblock-

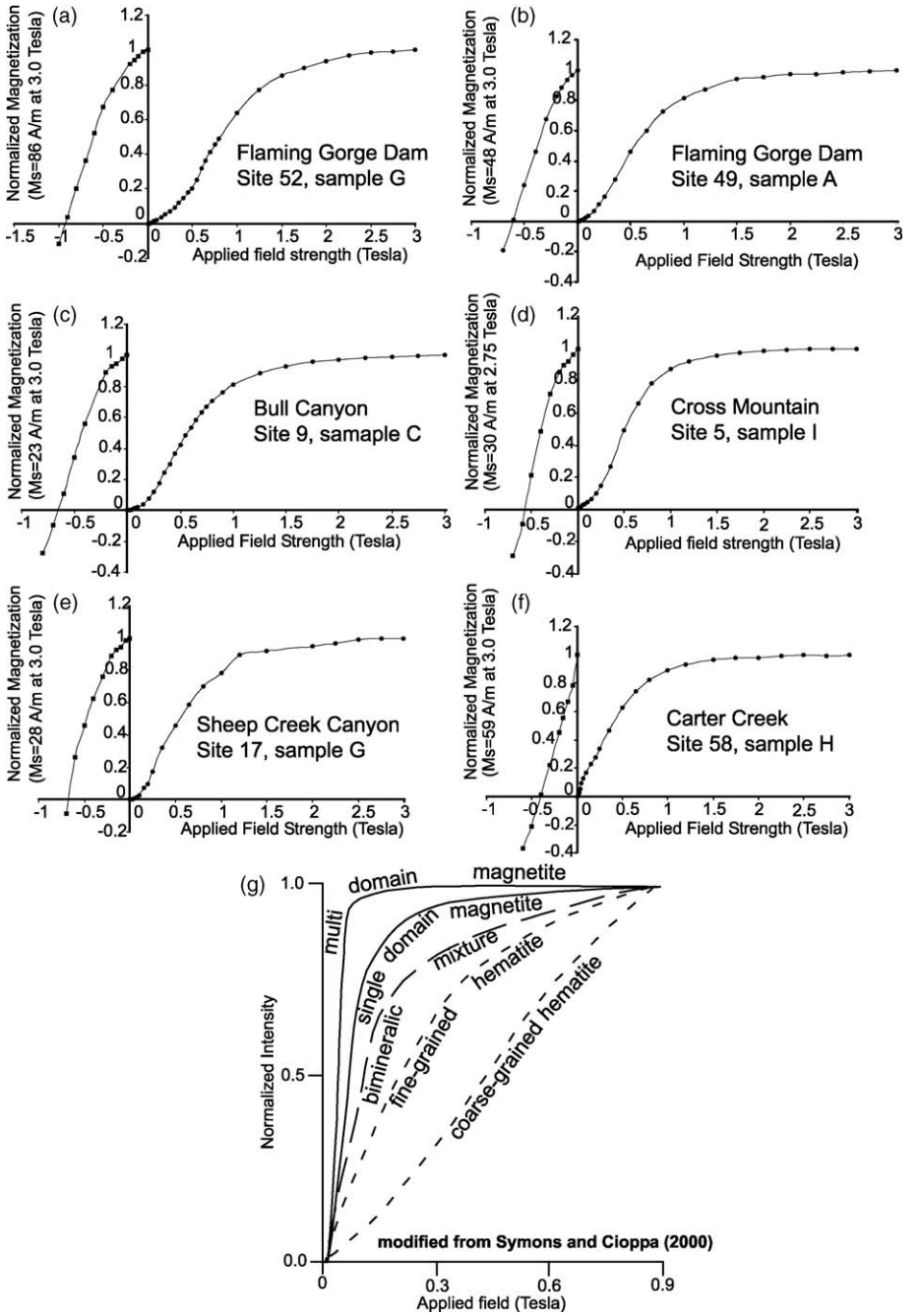


Fig. 9. (a–f) Graphs showing the acquisition of isothermal remanent magnetization (IRM) and subsequent, backfield demagnetization of saturation (or near-saturation) IRM for selected UMG rocks. (g) Typical (idealized) IRM acquisition curves for assemblages dominated by specific magnetic phases of generalized grain size. Note that the IRM acquisition curves for hematite and bimineralic mixtures are not saturated at 0.9 T (modified from Symons and Cioppa, 2000).

ing temperatures from about 660° to 680 °C. Samples from coarser-grained UMG rocks (about 0.25–1.5 mm) as well as ones from fine-grained yet relatively poorly indurated rocks typically exhibit more thermally distributed unblocking temperatures that range from about 600 to 660 °C. They also typically reveal early-removed to pervasive and complete north to north northeast-directed and positive inclination magnetizations.

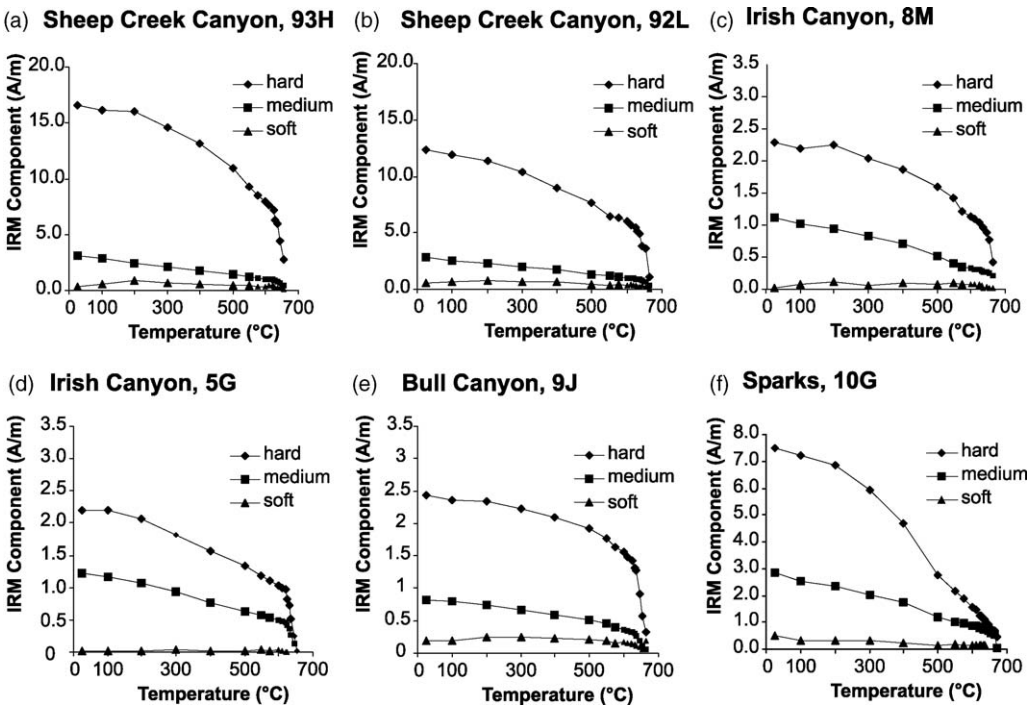


Fig. 10. (a–f) Diagrams showing the response to thermal demagnetization of three-component IRM, acquired in fields of 3.0 T ("hard" component), 0.3 T ("medium" component) and 0.03 T ("soft" component), following the approach of Lowrie (1990). In all examples, the hard component has the highest intensity, while the soft component has the lowest intensity.

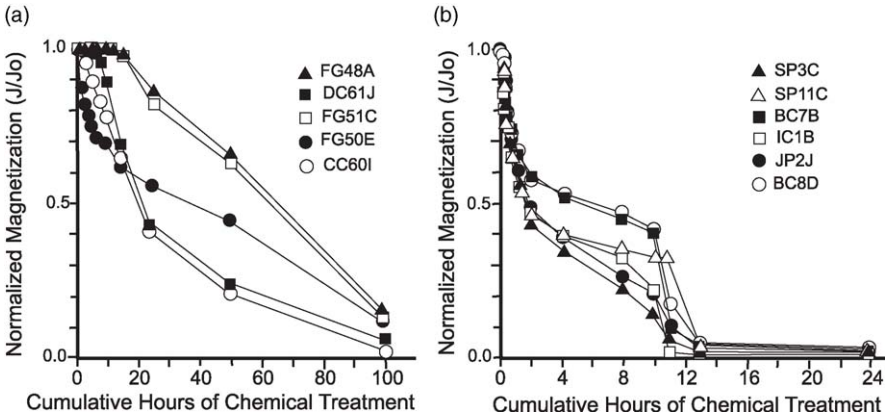


Fig. 11. (a and b) Normalized intensity vs. chemical demagnetization treatment (in cumulative hours of leaching in reagent HCl) for several samples from UMG strata. (a) Specimens from localities in the central part of the Uinta Mountains. (b) Specimens from localities in the eastern part of the Uinta Mountains.

5. Discussion

Thermal demagnetization results and rock magnetic experiments reveal the presence of two discrete magnetizations in UMG rocks, both of which are carried by hematite and are present at individual UMG sites, as well as at the locality level. The dominant ChRM is an east-directed (or antipode) shallow, positive to negative inclination magnetization that is typically unblocked over a narrow range of high laboratory temperatures between 660 and 680 °C (Fig. 6). The second magnetization is north to northeast-directed and has a moderate to steep, positive inclination (Fig. 5). The response of UMG samples to thermal and chemical demagnetization, comparison with previously obtained data from UMG rocks in the area, results from rock magnetic experiments, and the inferred, low-latitude of Laurentia during UMG deposition in the Neoproterozoic are

all consistent with the interpretation that the west- or east-directed, shallow magnetization is a near primary remanence. We recognize the fact that the direction of this magnetization is similar to, and thus yields paleomagnetic poles similar to, primary magnetizations from Late Cambrian to earliest Ordovician rocks for North America (e.g., Van der Voo, 1993). Although we have no direct, field-based proof of this, we argue that the ChRM in the UMG is not of early Phanerozoic age. The presence of dual polarities of the ChRM, with the west-directed remanence inferred to be of normal polarity and the east-directed of reverse polarity, suggests that magnetization acquisition in the UMG rocks took place over a sufficient amount of time to average secular variation of the geomagnetic field. We note, however, that, with the exception of the Carter Creek and Dowd Springs localities where several sites have mixed polarities, each locality was characterized by only one polarity. At this point, we cannot begin to establish a polarity stratigraphy for UMG strata in the central and eastern Uinta Mountains, although we note that very thick sections (e.g., Sheep Creek locality) may have been deposited during a single polarity chron. The paleomagnetic poles (mean of 0.8°N, 161.3°E, $\alpha_{95} = 4.6°$, $N = 9$ sampling localities consisting of 79 sites) derived from this inferred primary magnetization are fully consistent with recently reported results from rocks of mid-Neoproterozoic age for southwest Laurentia (Fig. 12) (Weil et al., 2003, 2004) as well as Neoproterozoic rocks from elsewhere in Laurentia that have been dated by isotopic methods (e.g., Park et al., 1995).

The north to northeast-directed, positive inclination, normal polarity magnetization is pervasive at several UMG sampling localities and is superimposed on, and often readily isolated from, the inferred primary magnetization identified at other localities. This magnetization is secondary and is likely very young on the basis of its direction and post-tilting (Laramide age) acquisition—it cannot be older than the mid-Cretaceous. This is contrary to numerous sedimentary sequences of pre-late Paleozoic age in the Rocky Mountain area, which have experienced earliest Pennsylvanian to mid-Permian aged remagnetization (summarized by Geissman and Harlan, 2002 and references therein).

For UMG rocks, grain size and degree of induration seem to be factors contributing to whether a specific locality or specific beds were partly to completely remagnetized. Samples that retain Neoproterozoic magnetizations are typically fine-grained (about 0.075–0.25 mm) and well-cemented with silica, and rocks dominated by northeast to north-directed positive inclination magnetization are coarser in grain size (about 0.25–1.5 mm).

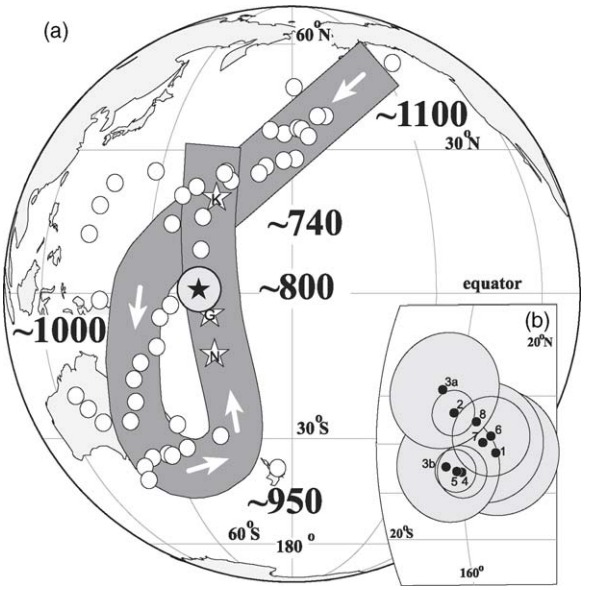


Fig. 12. (a) Summary apparent polar wander path for Laurentia (orthographic projection) for the 1100–700 Ma time interval. Approximate ages are given in millions of years (Ma). Open circles are selected Keweenawan and Grenville "A poles" from Weil et al. (1998) and Warnock et al. (2000). Stars represent recently published Neoproterozoic poles from the Grand Canyon Supergroup, Arizona; K, Kwagunt Formation; G, Galeros Formation; N, Nankoweap Formation (see Weil et al., 2003, 2004). (b) Close-up of the pole locations for individual sampling localities from the UMG (data taken from Table 3). (1) Browne Lake, (2) Sheep Creek Canyon, (3) Down Springs (a-flat lying strata; b-folded strata), (4) Carter Creek, (5) Flaming Gorge Dam, (6) Sparks–Talamantes Creek, (7) Home Mountain Canyon, (8) Bull Canyon.

Most of the magnetization in UMG samples is carried by fine-grained pigment hematite, a mineral phase usually deposited from solution (e.g., Turner, 1980; Larson and Walker, 1982). It is possible that, as a result of the enhanced porosity and permeability, coarser-grained rocks were remagnetized in response to fluid flow long after deposition of the UMG, overprinting, and at times and through specific sections, completely erasing the primary magnetization. One plausible remagnetization mechanism involves hydration of early-formed pigment hematite (authigenic) to goethite, and then subsequent goethite dehydration to a younger generation of hematite (Turner, 1980). However, many fine-grained UMG samples at both the site and entire locality level also contain the secondary magnetization and thus grain size, alone, cannot be the key determining factor in remagnetization. Pervasive remagnetization of hematite-dominated rocks from different geologic time periods has been observed in many studies (e.g., Peck et al., 1986; Miller and Kent, 1988; McCabe and Elmore, 1989; Weil et al., 2004) and must involve some form of reconstitution of

early-formed, authigenic hematite, although the process or processes that control this are not well understood.

Subsequent to mid to late Neoproterozoic deposition of UMG strata, the region remained essentially undeformed until latest Cretaceous to early Tertiary (ca. 70–35 Ma) time, which may explain the absence of secondary magnetizations of late Paleozoic age in these rocks. Deformation and the associated thermal enhancement and increased fluid flow in the crust are potential facilitators for remagnetization (e.g., Oliver, 1986; McCabe and Elmore, 1989; Geissman and Harlan, 2002). Initial faulting of the Uinta Mountains is not likely to have begun before about 70 Ma (Hansen, 1984; Bryant et al., 1989; Obradovich, 1993). Notably, paleomagnetic poles calculated from *in situ* directions of the secondary magnetization lie to the east, by up to about 40°, of those for North America for the Late Cretaceous to the present. The most extreme example of this is for the Sheep Creek Canyon locality, where 24 sites provide a mean of, in *in situ* coordinates, Decl. = 40.2°, Incl. = 63.6°, α_{95} = 5.5°, and pole at 62.2°N, 43.7°W (Fig. 5a), The primary to near primary magnetization recovered at this locality is exclusively of reverse polarity (east-directed and shallow) and thus it is possible that the NE declination of secondary magnetization simply represents a vector composite that progressive demagnetization is not capable of identifying. In addition, given the observed inclinations of the secondary magnetizations, the paleomagnetic poles derived from these magnetizations are not consistent with poles for time periods *older* than about 70 Ma (Van der Voo, 1993; Torsvik et al., 2001b; Schettino, 2001). The fact that the secondary magnetization in UMG strata is exclusively of normal polarity does not imply that acquisition was during a single normal polarity chron, although normal and reverse polarity chron durations during the latest Cretaceous to early Tertiary are relatively long (Cande and Kent, 1995). Several paleomagnetic studies of Mesozoic and older strata in the western Cordillera have reported young secondary magnetizations of a single polarity (e.g., McWinnie et al., 1990; Enkin et al., 2000; O'Brien et al., 2004). The acquisition of a chemical remanent magnetization (CRM) has been shown by Cairanne et al. (2003) to be ineffective in recording a dual polarity field over the time period of CRM acquisition. We suggest, but cannot conclusively demonstrate, that the secondary magnetization is of early Tertiary age, acquired during the waning stages of crustal shortening in the area. We refrain from considering the importance of the secondary magnetizations (Tertiary overprint directions), which are exclusively north to northeast-directed and of positive inclination, in the context of vertical axis rotation of any parts of the Uinta Mountains. This is largely because most of the data from the Uintas that convincingly demonstrate a pervasive secondary magnetization have magnetization directions that imply little if any statistically significant rotation, even if we assumed the secondary magnetization was as old as mid-Cretaceous. The notable exceptions to this are are the Sheep Creek (many sites) and Bull Canyon (one site) localities, where the inferred young, secondary declinations are, in *in situ* coordinates, oriented northeast to east northeast. We suspect, but cannot at this point prove, that this magnetization is actually a composite of the ChRM and a young, syn-to post Laramide deformation secondary magnetization. We do not think that arguments for substantial vertical axis rotation of this part of the Uinta Mountains, based on this magnetization, are valid.

At UMG localities where the inferred early-acquired, Neoproterozoic magnetization has been isolated, the paleomagnetic poles derived from all group mean data lie at low, equatorial latitudes, from about 6°N to 11°S and longitudes from 165.6°E to 155.6°E (Fig. 12; Table 3). These new poles are consistent with Bresler (1981) pole of 0.6°N, 157.2°E, based on data from sites at five separate localities. The pole data are also consistent with a recent re-interpretation of the Neoproterozoic APW path for Laurentia that shows a north-moving and slightly west of north track that crosses the equator near 160°E at about 800 Ma (Weil et al., 2003, 2004) (Fig. 12).

We cannot estimate the total amount of time during the Neoproterozoic spanned by the ensemble of ChRM data from the UMG. However, we note that the data from the UMG (Fig. 12) show little evidence for substantial APW and are perhaps inconsistent with a recent hypothesis of true polar wander (TPW) between about 800 and 750 Ma that is linked to a Rodinia superplume and eventual Rodinia dispersal (Li and Evans, 2004).

At present the best age-estimates for UMG rocks, based on paleontologic evidence and recent U/Th–Pb microprobe dating of early diagenetic monazites is 800–750 Ma. When compared to other recently obtained poles from southwest Laurentia, the most internally consistent APWP for the Neoproterozoic is a counterclockwise path that passes down through the south Pacific (Fig. 12). The only well-dated rocks of this time period that have yielded high-quality paleomagnetic data are the mafic igneous rocks of the Gunbarrel magmatic event (e.g., Harlan et al., 2003) that are dated by U–pb baddeleyite methods to yield a composite U–pb concordia age of 780.3 ± 1.4 Ma (95% confidence level). Overall, the paleomagnetic data from these rocks compare favorably with those from the UMG, but some of the well-dated paleomagnetic results are from single intrusions, and

there is real possibility that single intrusions have not done an adequate job of averaging paleosecular variation. The Concajou Canyon gabbro of the Tsezotene sills in the northern Cordillera (Northwest Territories, Yukon, British Columbia) yields an independent $^{207}Pb/^{206}Pb$ baddeleyite age determination of 779 ± 2.3 Ma (Harlan et al., 2003); the composite Tsezotene sill paleomagnetic pole is 1.6° N, 137°.8E ($\alpha_{95} = 5.0°$) (Park and Aitken, 1986). The Christmas Lake dike (Beartooth Mountains) is dated at 779.5 ± 3.0 Ma ($^{207}Pb/^{206}Pb$ baddeleyite methods), which is similar to the 777 ± 4 Ma $^{40}Ar/^{39}Ar$ hornblende plateau date of Harlan et al. (1997) and yields a pole at 7.4°N, 137.0°E (dp = 4.4°, dm = 8.3°) (Harlan et al., 1997). Finally, the Mount Moran dike (Teton Range), which is interpreted to be essentially identical in age to the Christmas Lake dike ($^{40}Ar/^{39}Ar$ hornblende plateau data of ca. 772 Ma., Harlan et al., 1997), yields a pole at 11.6°N, 148.9°E (dp = 5.1°, dm = 10.2°) (Harlan et al., 1997). The paleomagnetic results from the Franklin dikes, dated at aboiut 723 Ma, also yield similar results (pole at 9.2°N, 153.4°E) (Christie and Fahrig, 1983) are also very similar to 780 Ma data and those from the UMG. It should be noted that the segment of Laurentia's path between 1000 and 900 Ma is still ambiguous, due to the lack of undeformed rocks and the effects of Grenville metamorphism during this time. Comparing our UMG results with recently determined poles of similar age from several East Gondwana blocks and South China (Evans et al., 2000; Torsvik et al., 2001a; Torsvik et al., 2001c; Wingate and Giddings, 2000; Torsvik, 2003), it is clear that the present-day Cordilleran margin of Laurentia had begun its rift-to-drift transition prior to about 750 Ma, although the details of the dispersal of its neighbors during the Cryogenian and Ediacaran periods remain elusive.

6. Conclusion

The antiquity of paleomagnetic data obtained from UMG strata is partly demonstrated through the similarity of results from numerous geographically distinct sampling localities. Rock magnetic experiments show that hematite is the principal carrier of both a near primary and a secondary magnetization in these rocks. The persistence to high laboratory unblocking temperatures, the presence of dual-polarity magnetizations, as well as well-defined response to chemical demagnetization of the west-directed and shallow negative to positive inclination magnetization (and its antipode) support our interpretation that this ChRM in UMG rocks was acquired contemporaneous with or soon after UMG deposition in the mid to late Neoproterozoic. A north to northwest-directed, positive inclination magnetization is pervasive at some localities and is interpreted as secondary in origin and probably latest Cretaceous or younger in age, possibly acquired during Laramide age deformation of the Uinta Mountains.

The primary ChRM from the UMG is consistent with a tight counterclockwise APWP loop that passes down through the south Pacific for Laurentia during the Neoproterozoic, which is in contrast to recently published clockwise APWP loops for Laurentia for the 1100–700 Ma time interval (e.g., Meert and Torsvik, 2003 and references therein). UMG data are of about the same age as recently published data from East Gondwana, which when combined with additional recent data from the southwestern U.S. (Weil et al., 2003, 2004) suggests that the breakup of the proposed Rodinia supercontinent had begun before 750 Ma. The APWP (Fig. 12) suggests that Laurentia drifted northward from high southern latitudes to equatorial latitudes during the 950–750 Ma time interval. This drift history positioned the supercontinent Laurentia at or near the equator between ~800 and ~740 Ma, consistent with the hypothesized Sturtian low-latitude snowball Earth event (Hoffman et al., 1998a,b; Hoffman and Schrag, 2002; Fanning and Link, 2004).

Q&A section

Harlan: Although this work is up to date and significantly more reliable than previous work, how different is the mean Uinta pole reported here from the previous pole published by Bresler et al.?

In the grand scheme of things, the mean Uinta pole reported here differs slightly from that reported by Bresler (1981) (not Bresler et al., 1981). Bresler (1981) reported collecting UMG strata from seven localities, four of which lie farther to the west in the Uinta Mountains than any of our localities. Ultimately, he combined the results from two eastern localities, and the data were considered as, overall, two results, one from his locality 6, which was the highest UMG strata sampled in the western part of the range, and one that combined results from localities one through five. His localities one through five all yielded west-directed and shallow negative inclination directions, and a pole of 0.6°N/157.2°E. This result is slightly east of and shallower in latitude than our grand mean result, but not statistically distinguishable. The result from his locality 6, however, is based on 13 samples, all of which showed an east-directed magnetization and resulted in a pole (4.6°N, 140.1°E) that lies some 20°E of our grand mean. We

argue that our result provides a more robust sampling of the field during this part of the Neoproterozoic.

McCausland: Establishing the 'primary' nature of the hematite pigment-borne ChRM is very important! As you note, the pole position could either be mid-Neoproterozoic or Cambrian. What might be done to constrain the age of hematite pigment formation? Is there a spatial field relationship between the silica-cemented sites which retain the ancient remanence and the mid-Cambrian angular unconformity?

This is an excellent question that certainly deserves further attention. Notably, in most parts of the Uinta Mountains, UMG strata are in fault contact with Phanerozoic rocks. Exceptions exist along the southern flank of the range, but in these areas the disconformity between UMG and Cambrian strata is poorly exposed. In Irish Canyon, however, the disconformity is well exposed and we conducted extensive sampling of UMG strata both immediately beneath as well as about 500 m below the disconformity. We report data from UMG rocks about 500 m below the disconformity, which are completely remagnetized. All of the UMG strata immediately below the disconformity are completely remagnetized as well; these data are not reported here.

Acknowledgments

The following individuals assisted us in sampling in the Uinta Mountains: Carol M. Dehler, Amanda Rogers, Rick Ortiz, Doug Raszewski and Chris Purcell. Mike Petronis is acknowledged for assistance in the UNM laboratory. We thank the Ashley National Forest for our research agreement and sampling permit. We acknowledge several sources of funding for this work, including Frank Gorham, Jr., The Four Corners Geologic Society, The Geologic Society of America, The University of New Mexico, and the Department of Earth and Planetary Sciences. Carol Dehler is also thanked for discussion and comments on an earlier draft of this manuscript. Reviewers Stephen Harlan and Trond Torsvik, and co-editor Phil McCausland are thanked for their helpful comments, which greatly improved the quality of this manuscript.

References

Alvarez, V.C., Dunlop, D.J., 1998. A regional paleomagnetic study of lithotectonic domains in the Central Gneiss Belt, Grenville Province, Ontario. Earth Planet. Sci. Lett. 157, 89–103.

Berger, G.W., York, D., Dunlop, D.J., 1979. Calibration of Grenvillian palaeopoles by $^{40}Ar/^{39}Ar$ dating. Nature 277, 46–47.

Berger, G.W., York, D., 1981. $^{40}Ar/^{39}Ar$ dating of the Thanet gabbro, Ontario: looking through the Grenvillian metamorphic veil and implications for paleomagnetism. Can. J. Earth Sci. 18, 266–273.

Bird, P., 1998. Kinematic history of the Laramide orogeny in latitudes 35°–49°, western United States. Tectonics 17, 781–801.

Bond, G.C., Kominz, M.A., 1984. Construction of tectonic subsidence curves for the early Paleozoic miogeocline, southern Canadian Rocky Mountains: Implications for subsidence mechanisms, age of break-up, and crustal thinning. Geol. Soc.Am. Bull. 95, 155–217.

Bresler, S.L., 1981. Preliminary paleomagnetic poles and correlation of the Proterozoic Uinta Mountain Group, Utah and Colorado. Earth Planet. Sci. Lett. 55, 53–64.

Bryant, B., Naeser, C.W., Marvin, R.F., Mehnert, H.H., 1989. Upper Cretaceous and Paleogene sedimentary rocks and isotopic ages of Paleogene tuffs, Uinta Basin, Utah. Ages of late Paleogene and Neogene tuffs and the beginning of rapid regional extension, eastern boundary of the basin and range province near Salt Lake City, Utah: evolution of sedimentary basins, Uinta and Piceance basins, U.S. Geol. Surv. Bull. J1–J22, K1–K12.

Bryant, B., Nichols, D.J., 1988. Late Mesozoic and early Tertiary reactivation of an ancient crustal boundary along the Uinta trend and its interactions with the Sevier orogenic belt. Memoir—Geol. Soc. Am. 171, 411–430.

Buchan, K.L., Mertanen, S., Pesonen, L.J., Elming, S.A., Abrahamsen, N., Bylund, G., 2000. Comparing the drift of Laurentia and Baltica in the Proterozoic: the importance of key palaeomagnetic poles. Tectonophysics 319, 167–198.

Buchan, K.L., Ernst, R.E., Hamilton, M.A., Mertanen, S., Pesonen, L.J., Elming, S., 2001. Rodinia: the evidence from integrated palaeomagnetism and U–Pb geochronology. Precambrian Res. 110, 9–32.

Cairanne, G., Brunet, F., Pozzi, J.P., Besson, P., Aubourg, C., 2003. Magnetic monitoring of hydrothermal magnetite nucleation-and-growth: record of magnetic reversals. Am. Mineral. 88, 1385–1389.

Cande, S.C., Kent, D.V., 1995. Revised calibration of the geomagnetic polarity timescale for the Late Cretaceous and Cenozoic. J. Geophys. Res. 100, 6093–6095.

Christie, K.W., Fahrig, W.F., 1983. Paleomagnetism of the Borden dykes of Baffin Island and its bearing on the Grenville Loop. Can. J. Earth Sci. 20, 275–289.

Collinson, D.W., 1966. Carrier of remanent magnetization in certain red sandstones. Nature 210, 516–517.

Collinson, D.W., 1974. The role of pigment and specularite in the remanent magnetism of red sandstones. Geophys. J. R. Astronom. Soc. 38, 253–264.

Condie, K.C., Dennis, L., Farmer, G.L., 2001. Tectonic setting and provenance of the Neoproterozoic Uinta Mountain and Big Cottonwood groups, northern Utah; constraints from geochemistry, Nd isotopes, and detrital modes. In: Eriksson, P.G., Catuneanu, O., Aspler, L.B., Chiarenzelli, J.R. (Eds.), The influence of magmatism, tectonics sea level change and paleo-climate on Precambrian basin evolution; change over time Martins, vol. 141–142. Sedimentary Geology, pp. 443–464.

Dalziel, I.W.D., 1992. On the organization of American plates in the Neoproterozoic and the breakout of Laurentia. GSA Today 2 (11), 237–241.

Dehler, C.M., Des Marias, D.J., Bowring, S., Sharp, Z., Karlstrom, K.E., 2001a. Elrick Neoproterozoic Chuar Group (~800–742 Ma), Grand Canyon: a record of cyclic marine deposition during global cooling and supercontinent rifting. Sediment. Geol. 141–142, 465–499.

Dehler, C.M., Prave, A., Crossey, L.J., Karlstrom, K.E., 2001b. Linking Mid-Neoproterozoic successions in the western United States: The Chuar Group-Uinta Mountain Group-Pahrump Group connection (CHUMP). Geological Society of America, Abstracts with Programs, South-Central/Rocky Mountain Sectional Meeting. Albuquerque, New Mexico 33, A20–A21.

Dehler, C.M., Porter, S.M., Sprinkel, D.A., DeGrey, L.D. Waanders, G., 2006. The Neoproterozoic Uinta Mountain Group revisited: a synthesis of recent work on the Red Pine Shale and undivided clastic strata, northeastern Utah. In: Link, P.K., Lewis, R., (Eds.), Proterozoic Geology of the Northwest U.S.: Basement and Tectonic Setting of the Belt Basin and Succeeding Windermere Supergroup: Society for Sedimentary Geology (SEPM).

Dunlop, D.J., 1972. Magnetic mineralogy of unheated and heated red sediments by coercivity spectrum analysis. Geophys.J. R. Astronom. Soc. 27, 37–55.

Dunlop, D.J., Ozdemir, O., 1997. Rock magnetism; fundamentals and frontiers. Cambridge University Press, Cambridge, United Kingdom, p. 573.

Elston, D.P., 1993. Middle and Early–late Proterozoic Grand Canyon Supergroup, northern Arizona. In: Reed, J.C., Bickford, M.E., Houston, R.S., Link, P.K., Rankin, D.W., Sims, P.K., Van Schmus, W.R. (Eds.), Precambrian, Conterminous U.S.: Boulder, Colorado, vol. C-2. Geological Society of America, Geology of North America, pp. 521–529.

Enkin, R.J., Osadetz, K.G., Baker, J., Kisilevsky, D., 2000. Orogenic remagnetizations in the Front Range and Inner Foothills of the southern Canadian Cordillera: Chemical harbinger and thermal handmaiden of Cordilleran deformation. Geologic. Soc. Am. Bull. 112, 929–942.

Evans, D.A.D., Li, Z.X., Kirschvink, J.L., Wingate, M.T.D., 2000. A high-quality mid-Neoproterozoic paleomagnetic pole from South China, with implications for ice ages and the breakup configuration of Rodinia. Precambrian Res. 100, 313–334.

Fanning, C.M., Link, P.K., 2004. U–pb Shrimp age of Neoproterozoic (Sturtian) glaciogenic Pocatello Formation, southeastern Idaho. Geology 32, 881–884.

Farmer, G.L., Ball, T.T., 1997. Sources of Middle Proterozoic to Early Cambrian siliciclastic sedimentary rocks in the Great Basin: a Nd isotope study. Geologic. Soc. Am. Bull. 109, 1193–1205.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. Lond. A 217, 295–305.

Geissman, J.W., Harlan, S.S., 2002. Late Paleozoic remagnetization of Precambrian crystalline rocks along the Precambrian/Carboniferous nonconformity, Rocky Mountains: a relationship among deformation, remagnetization, and fluid migration. Earth Planet. Sci. Lett. 203, 905–924.

Gradstein, F.M., Ogg, J.G., Smith, A.M., 2004. A Geologic Time Scale. Cambridge University Press, Cambridge, p. 610.

Gries, R.R., 1983. Oil and gas prospecting beneath Precambrian of foreland thrust plates in Rocky Mountains. Am. Assoc. Petroleum Geol. Bull. 67, 1–28.

Halls, H.C., Pesonen, L.J., 1982. Paleomagnetism of Keweenawan rocks. Geol. Soc. Am. Memoir. 156, 173–201.

Hansen, W.R., 1965. Geology of the Flaming Gorge area, Utah-Colorado-Wyoming. U.S. Geological Survey Professional Paper, vol. 490, p. 196.

Hansen, W.R., 1984. Post-Laramide tectonic history of the eastern Uinta Mountains, Utah, Wyoming, Colorado. The Mountain Geol. 21, 5–29.

Harlan, S.S., Geissman, J.W., Snee, L.W., 1997. Paleomagnetism and $^{40}Ar/^{39}Ar$ geochronologic data from late Proterozoic mafic dikes and sills, Montana and Wyoming. U. S. Geol.Surv. Professional Paper, 16.

Harland, W.B., 1964. Critical evidence for a great infra-Cambrian glaciation. Geol. Rundsch. 54, 45–61.

Harlan, S.S., Heaman, L., LeCheminant, A.N., Premo, W.R., 2003. Gunbarrel mafic magmatic event; a key 780 Ma time marker for Rodinia plate reconstructions. Geology 31, 1053–1056.

Harrison, J.E., Griggs, A.B., Wells, J.D., 1974. Tectonic features of the Precambrian Belt basin and their influence on post-Belt structures. U. S. Geol. Surv. Professional Paper 886, 15.

Henry, S.G., 1979. Chemical demagnetization; methods, procedures, and applications through vector analysis. Can. J. Earth Sci. 16, 1832–1841.

Hoffman, P.F., 1991. Did the breakout of Laurentia turn Gondwana inside out? Science 252, 1409–1412.

Hoffman, P.F., Kaufman, A.J., Halverson, G.P., 1998a. Comings and goings of global glaciations in a Neoproterozoic tropical platform of Namibia. GSA Today 8 (5), 1–9.

Hoffman, P.F., Kaufman, A.K., Halverson, G.P., Schrag, D.P., 1998b. A Neoproterozoic Snowball Earth. Science 281, 1342–1346.

Hoffman, P.F., 1999. The breakup of Rodinia, birth of Gondwana, true polar wander, and the snowball Earth. J. Afr. Earth Sci. 28, 9–26.

Hoffman, P.F., Schrag, D.P., 2002. The snowball Earth hypothesis: testing the limits of global change. Terra Nova 14, 129–155.

Hyodo, H., Dunlop, D.J., 1993. Effect of Anisotropy on the paleomagnetic contact test for a Grenville dike. J. Geophys. Res. 98, 7887–8017.

Irving, E., 1979. Paleopoles and paleolatitudes of North America and speculations about displaced terraines, In: Price, R. A. (Ed.), Geodynamics Canada, Can. J. Earth Sci. 16, 669–694.

Karlstrom, Karl, E., Bowring, S.A., Dehler, C.M., Knoll, A.H., Porter, S.M., Sharp, Z., Des Marais, D.J., Weil, A.B., Geissman, J.W., Elrick, M., Timmons, M.J., Keefe, K., Crossey, L.J., 2000. The Chuar Group of the Grand Canyon: Record of break up of Rodinia, associated change in the global carbon cycle, and eukaryotic diversification by 740 Ma. Geology 28, 619–622.

Kaufman, A.J., Knoll, A.H., 1995. Neoproterozoic variations in the C-isotope composition of seawater: Stratigraphic and biogeochemical implications. Precambrian Res. 73, 27–49.

Kaufman, A.J., 1997. An ice age in the tropics. Nature 386, 227–228.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophys. J. R. Astronom. Soc. 62, 699–718.

Kirschvink, J.L., 1992. Late Proterozoic low-latitude global glaciation. In: Schopf, J.W., Klein, C. (Eds.), The Snowball Earth The Proterozoic Biosphere. Cambridge University Press, Cambridge, pp. 51–52.

Knoll, A.H., 1992. The early evolution of eukaryotes: a geological perspective. Science 94, 6600–6605.

Larson, E.E., Walker, T.R., 1982. A rock magnetic study of the Lower Massive Sandstone, Moenkopi Formation (Triassic), Gray Mountain area, Arizona. J. Geophys. Res. 87, 4819–4836.

Li, Z.X., Evans, D.A.D., 2004. A 90 degrees spin on Rodinia: possible causal links between the Neoproterozoic supercontinent, superplume, true polar wander and low-latitude glaciation. Earth Planet. Sci. Lett. 220, 409–421.

Link, P.K., Christie-Blick, N., Devlin, W.J., Elston, D.P., Horodyski, R.J., Levy, M., Miller, J.M.G., Pearson, R.C., Prave, A., Stewart, J.H., Winston, D., Wright, L.A., and Wrucke, C.T., 1993. Middle and Late Proterozoic stratified rocks of the western U. S. Cordillera, Colorado Plateau, and Basin and Range Province, in: Reed, J.C., Bickford, M.E., Houston, R.S., Link, P.K., Rankin, D.W., Sims,

P.K., Van Schmus, W.R., eds., Precambrian: Coterminous U.S., Boulder, CO. Geological Society of America, The Geology of North America, C-2, pp. 463–595.

Lowrie, W., 1990. Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties. Geophys. Res. Lett. 17, 159–162.

Meert, J.G., 2003. A synopsis of events related to the assembly of eastern Gondwana. Tectonophysics 362, 1–40.

Meert, J.G., Powell, C.McA., 2002. Editorial: Assembly and break-up of Rodinia: introduction to the special volume. Precambrian Res. 110, 1–8.

Meert, J.G., Torsvik, T.H., 2003. The making and unmaking of a supercontinent: Rodinia revisited. Tectonophysics 375, 261–288.

McCabe, C., Elmore, R.D., 1989. The occurrence and origin of late Paleozoic remagnetization in the sedimentary rocks of North America. Rev. Geophys. 27, 471–494.

McFadden, P.L., McElhinny, M.W., 1988. The combined analysis of remagnetization circles and direct observations in palaeomagnetism. Earth Planet. Sci. Lett. 87, 161–172.

McWilliams, M.O., Dunlop, D.J., 1978. Grenville paleomagnetism and tectonics. Can. J. Earth Sci. 15, 687–695.

McWinnie, S.T., Van der Pluijm, B., Van der Voo, R., 1990. Remagnetization and thrusting in the Idaho-Wyoming overthrust belt. J. Geophys. Res. 95, 4551–4559.

Miller, J.D., Kent, D.V., 1988. Regional trends in the timing of Alleghenian remagnetization in the Appalachians. Geology 16, 588–591.

Obradovich, J.D., 1993. A Cretaceous time scale. In: Caldwell, W.G.E., Kauffman, E.G. (Eds.), Evolution of the Western Interior Basin. Geologic Association of Canada, Special Paper 39, 379–396.

O'Brien, V., Moreland, K.M., Elmore, R.D., Engel, M.H., Dulin, S.A., 2004. Orogenic remagnetizations, Sawtooth Mountains, Montana. Geol. Soc. Am. Abstr. Programs 36, 174.

Oliver, J., 1986. Fluids expelled tectonically from orogenic belts: Their role in hydrocarbon migration and other geologic phenomena. Geology 14, 99–102.

Park, J.K., 1994. Palaeomagnetic constraints on the position of Laurentia from middle Neoproterozoic to Early Cambrian times. Precambrian Res. 69, 95–112.

Park, J.K., Aitken, J.D., 1986. Paleomagnetism of the late Proterozoic Tsezotene Formation of northwestern Canada. J. Geophys. Res. 91, 4955–4970.

Park, J.K., Buchan, K.L., Gandhi, S.S., 1995. Paleomagnetism of 779 Ma Hottah gabbro sheets of the Wopmay Orogen Northwest Territories. Current Res.—Geol. Surv. Can., 195–200.

Peck, R.D., Elmore, R.D., Dubois, R.L., 1986. Early and late Paleozoic remagnetization of the Cambrian Peerless Formation, Colorado. Phys. Earth Planet. Interior 43, 274–282.

Pesonen, L.J., Elming, S.-Å., Mertanen, S., Pisarevsky, S., D'Argrella-Filho, M.S., Meert, J.G., Schmidt, P.W., Abrahamsen, N., Bylund, G., 2003. Palaeomagnetic configuration of continents during the Proterozoic. Tectonophysics 375, 289–324.

Pisarevsky, S.A., Wingate, M.T.D., Powell, C.McA., Johnson, S., Evans, D.A., 2003. Models of Rodinia assembly and fragmentation. In: Yoshida, M., Windley, B.F., Dasgupta, S. (Eds.), Proterozoic East Gondwana: Supercontinent Assembly and Breakup, vol. 206. Geologic Society, London, Special Publication, pp. 35–55.

Porter, S., Knoll, A.H., 2000. Testate amoebae in the Neoproterozoic Era: evidence from vase-shaped microfossils in the Chuar Group, Grand Canyon. Paleobiology 26, 345–370.

Powell, C.McA., Li, Z.X., McElhinny, M.W., Meert, J.G., Park, J.K., 1993. Paleomagnetic constraints on timing of the Neoproterozoic breakup of Rodinia and the Cambrian formation of Gondwana. Geology 21, 889–892.

Roy, J.L., 1983. Paleomagnetism of the North American Precambrian: a look at the data base. Precambrian Res. 19, 319–348.

Sanderson, I.D., 1984. The Mount Watson Formation, an interpreted braided-fluvial deposit in the Uinta Mountain Group (upper Precambrian), Utah. The Mountain Geol. 21, 157–164.

Sanderson, I.D., Wiley, M.T., 1986. The Jesse Ewing Canyon Formation, an interpreted alluvial fan deposit in the basal Uinta Mountain Group (Middle Proterozoic), Utah. The Mountain Geol. 23 (3), 77–89.

Sears, J.W., Graff, P.J., Holden, G.S., 1982. Tectonic evolution of lower Proterozoic rocks, Uinta Mountains Utah and Colorado. Geol. Soc. Am. Bull. 93, 990–997.

Schell, E.M., 1969. Summary of the geology of Sheep Creek Canyon geological area and vicinity, Daggett County, Utah. In: Lindsay, J.B. (Ed.), Geologic Guidebook of the Uinta Mountains, Utah's Maverick Range: Intermountain Association of Petroleum Geologists Guidebook 16th Annual Field Conference, 143–152, pl. 1.

Schettino, A., 2001. The Global Paleomagnetic Data Base (GPDB) Internet Interface, http://www.itis-molinari.mi.it/pmag.htm.

Sprinkel, D.A., Park, B., Stevens, M., 2000. Geology of Sheep Creek Canyon geological area northeastern Utah. In: Sprinkel, D.A., Chidsey Jr., T.C., Anderson, P.B. (Eds.), Geology of Utah's parks and monuments, vol. 28. Utah Geological Association Publication, pp. 517–528.

Sprinkel, D.A., Waanders, G., Robbins, E.I., 2002. Stratigraphy, palynology, and maturity of the Proterozoic Uinta Mountain Group, eastern Uinta Mountains, Utah: implications for unit thickness. Geol. Soc. Am., Abstr. Programs, Rocky Mountain Sectional Meeting, vol. 34, p.18.

Stone, D.S., 1995. Tectonic evolution of the Uinta Mountains: Palinspastic restoration of a structural cross section along longitude 109°15′, Utah. Utah Geol. Surv. 23, 447–450.

Stone, D.S., 1993. Tectonic evolution of the Uinta Mountains: Palinspastic restoration of a structural cross section along longitude 109°15′, Utah. Utah Geological Survey, Miscellaneous Publication, 93–98.

Symons, D.T.A., Cioppa, M.T., 2000. Crossover Plots: a useful method for plotting SIRM data in paleomagnetism. Geophys. Res. Lett. 27, 1779–1782.

Torsvik, T.H., Carter, L.M., Ashwal, L.D., Bhushan, S.K., Pandit, M.K., Jamtveit, B., 2001a. Rodinia refined or obscured: palaeomagnetism of the Malani igneous suite (NW India). Precambrian Res. 108, 319–333.

Torsvik, T.H., Van der Voo, Rob, Meert, J.G., Mosar, J., Walderhaug, H.J., 2001b. Reconstructions of the continents around the North Atlantic at about the 60th parallel. Earth Planet. Sci. Lett. 187, 55–69.

Torsvik, T.H., Ashwal, L.D., Tucker, R.D., Eide, E.A., 2001c. Neoproterozoic geochronology and palaeogeography of the Seychelles microcontinent: the India link. Precambrian Res. 110, 47–59.

Torsvik, T.H., Smethurst, M.A., Meert, J.G., Van der Voo, R., McKerrow, W.S., Braiser, M.D., Sturt, B.A., Walderhaug, H.J., 1996. Continental break-up in the Neoproterozoic and Paleozoic-A tale of Baltica and Laurentia. Earth Sci. Rev. 40, 229–258.

Torsvik, T.H., 2003. The Rodinia Jigsaw Puzzle. Science 300, 1379–1381.

Turner, P., 1980. Continental Red Beds. Elsevier, Amsterdam, p. 562.

Van der Voo, R., 1993. Paleomagnetism of the Atlantic Tethys and Iapetus oceans. Cambridge University Press, Cambridge, United Kingdom (GBR), p. 411.

Vidal, G., Ford, T.D., 1985. Microbiotas from the Late Proterozoic Chuar Group (northern Arizona) and Uinta Mountain Group (Utah) and their chronostratigraphic implications. Precambrian Res. 28, 349–389.

Walderhaug, H.J., Torsvik, T.H., Eide, E.A., Sundvoll, B., Bingen, B., 1999. Geochronology and paleomagnetism of the Hunnedalen dykes, SW Norway: implications for the Sveconorwegian apparent polar wander loop. Earth Planet. Sci. Lett. 169, 71–83.

Wallace, C.A. and Crittenden, M.D., 1969. The stratigraphy, depositional environment and correlation of the Precambrian Uinta Mountain Group, Western Uinta Mountains, Utah, in: Lindsay, J.B., ed., Geologic Guidebook of the Uinta Mountains, Utah's Maverick Range. International Association Geologists 16th Annual Field Conference, pp. 127–141.

Wallace, C. A., 1972. A basin analysis of the upper Precambrian Uinta Mountain Group, University of California, Santa Barbara. Ph.D. Thesis, pp. 412.

Warnockl, A.C., Kodama, K.P., Zeitler, P.K., 2000. Using thermochronology and low-temperature demagnetisation to accurately date Precambrian poles. J. Geophys. Res. 105, 19435–19453.

Weil, A.B., Van der Voo, R., Mac Niocaill, C., Meert, G.M., 1998. The Proterozoic supercontinent Rodinia: paleomagnetically derived reconstructions for the 1100 to 800 Ma interval. Earth Planet. Sci. Lett. 154, 13–24.

Weil, A.B., Geissman, J., Heizler, J., Van der Voo, R., 2003. A paleomagnetic investigation of Middle Proterozoic mafic intrusions and Upper Proterozoic redbeds from the Lower Grand Canyon Supergroup, Arizona. Tectonophysics 375, 199–220.

Weil, A.B., Geissman, J., Van der Voo, R., 2004. Paleomagnetism of the Neoproterozoic Chuar Group, Grand Canyon Supergroup, Arizona: implications for Laurentia's Neoproterozoic APWP and Rodinia break-up. Precambrian Res. 129, 71–92.

Williams, M.L., Crossey, L.J., Jercinovic, M.J., Bloch, J.D., Karlstrom, K.E., Dehler, C.M., Heizler, M.T., Bowring, S.A., Goncalves, P., 2003. Dating sedimentary sequences: in situ U/Th–Pb microprobe dating of early diagenetic monazite and Ar–Ar dating of marcasite nodules: case study from Neoproterozoic black shales in the southwest U.S., Abstracts with Programs. Geol. Soc. Am. 34, 595.

Wingate, M.T.D., Giddings, J.W., 2000. Age and paleomagnetism of the Mundine Well dyke swarm, Western Australia: Implications for an Australia-Laurentia connection at 755 Ma. Precambrian Res. 100, 335–357.

Wingate, M.T.D., Pisarevsky, S.A., Evans, A.D., 2002. Rodinia connections between Australia and Laurentia: no SWEAT, no AUSWUS? Terra Nova 14, 121–128.

Wingate, M.T.D., Evans, D.A., 2003. Palaeomagnetic constraints on the Proterozoic tectonic evolution of Australia. In: Yoshida, M., Windley, B.F., Dasgupta, S. (Eds.), Proterozoic East Gondwana: Supercontinent Assembly and Breakup. Geologic Society, vol. 206. Special Publication, London, pp. 77–91.

Zijderveld, J.D.A., 1967. AC demagnetization of rocks: Analysis of results. In: Collinson, D.W., Runcorn, S.K., Creer, K.M. (Eds.), Methods in Paleomagnetism. Elsevier, New York, pp. 254–286.